MAMMA ADA CENTRE

In Collaboration with FACM ITALY

Class 8

AI Innovating Today Empowering Tomorrow

GAIO Global AI Olympiad

Think | Explore | Create | A Better Tomorrow

1. AI Basics with Real Life
2. Communicate with AI
3. AI Applications
4. AI & Career Enhancement
5. AI Tools with Demos
6. Responsible AI & Ethics

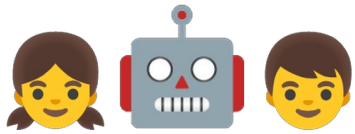
This book belongs to

Name: _____

Class & Section: _____

School: _____

📖 PREFACE



Artificial Intelligence is shaping how we learn, communicate, travel, farm, heal and create. As senior students, you are ready not only to use AI tools, but to understand how they work, question their answers and use them responsibly. This book is designed to build exactly those skills.

This book has been prepared for the Global AI Olympiad (GAIO) by Mamma Ada Centre, in collaboration with FACM, Italy. It follows the GAIO six-month journey: AI Discover, AI Connect, AI Solve, AI Rise, AI Create and AI Care. Each month introduces two topics suited to the curiosity and abilities of Class 8 learners.

Every topic opens with a clear introduction and continues with "Tech Talk", a group chat in which Riya, Kabir and their AI mentor Bolt explore ideas through real-life examples. A "How it works" flow, an India Spotlight, a hands-on Demo Lab with real AI tools, an "AI Chronicles" story, worksheets, quizzes, and standard and case-based exercises help you learn by reading, doing, questioning and creating.

Values are at the heart of this book. Along with learning what AI can do, children learn that people create and guide technology, that AI can make mistakes, and that safety, kindness and privacy matter online and offline.

🤝 A note for teachers and parents

Encourage students to try the Demo Labs under supervision, discuss the case studies, and question AI answers rather than accept them blindly. Praise reasoning and curiosity as much as right answers. An answer key is provided at the end of the book.

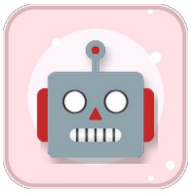
We hope this book sparks curiosity, creativity and confidence in every young learner. Curious minds, brighter futures!

With best wishes,

The GAIO Academic Team

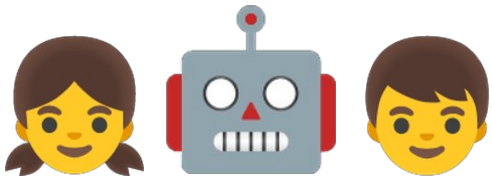
Mamma Ada Centre • In collaboration with FACM, Italy

Welcome, Seniors!



MEET BOLT!

Welcome, Class 8! I'm Bolt. This year: decision trees, machine learning, Python functions, GenAI projects, deepfakes and your data rights. Let's innovate!



Look for these pictures in your book:

TECH TALK Concepts in a group chat

AI CHRONICLES Stories of teens & AI

LOOK AROUND YOU See AI in real life

LET'S DO IT! Do a fun activity

WORKSHEET Write, tick and draw

PUZZLE FUN Solve puzzles

DEMO LAB Hands-on AI tool demo

QUIZ TIME Answer quiz questions

TRUE OR FALSE True or false?

EXERCISE Standard & case-based Qs

MAGIC WORDS New words to learn

UNIT REVIEW Check your unit

My 6-Month AI Journey

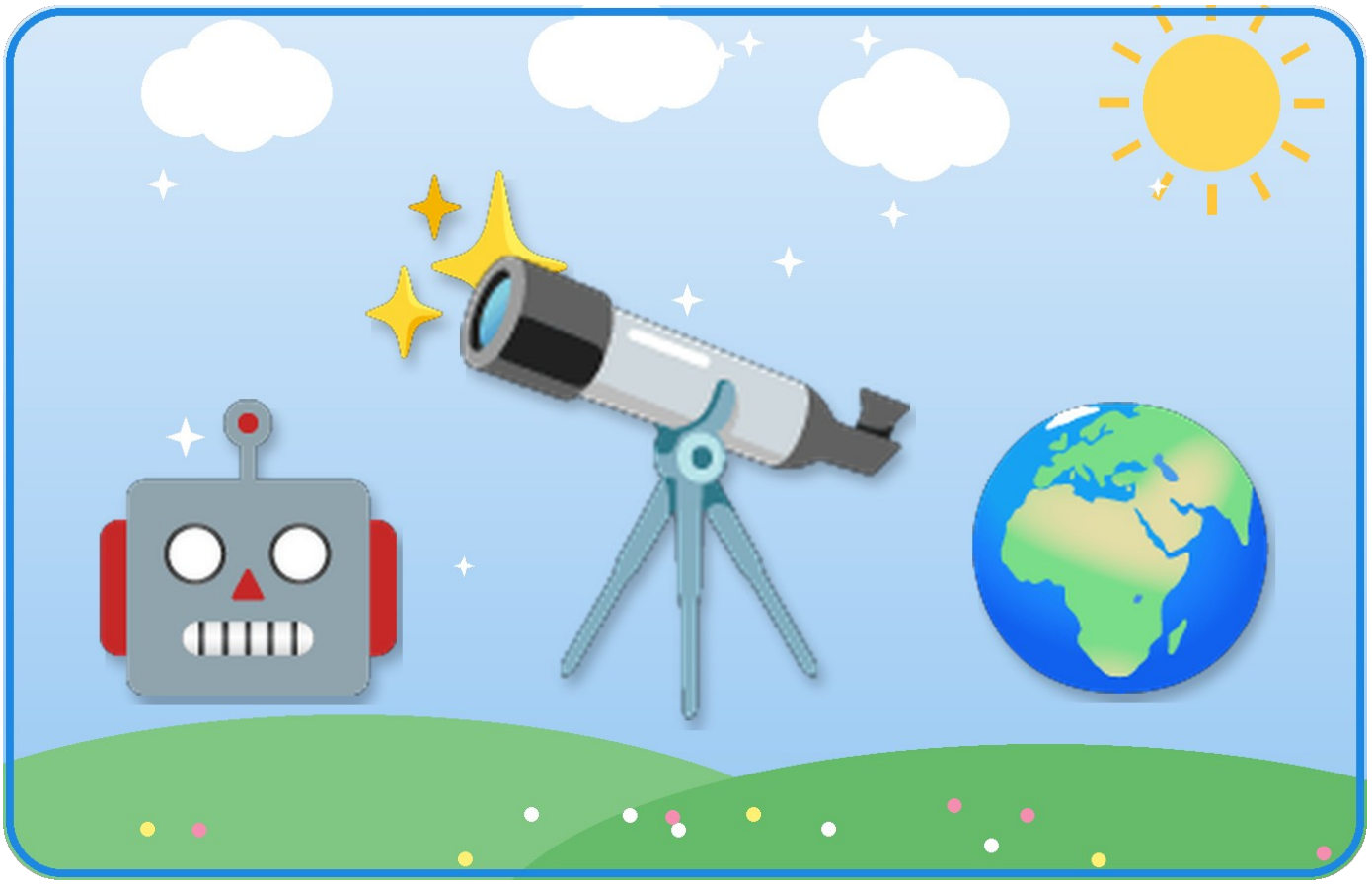
Follow the path. One month at a time!

MONTH 1	**AI DISCOVER** AI Basics	**How AI Makes Choices • Learning from Data**
MONTH 2	**AI CONNECT** Communicate with AI	**Python Playground • Code a Solution**
MONTH 3	**AI SOLVE** AI Applications	**AI for Better Health • AI for Better Cities**
MONTH 4	**AI RISE** AI & Career	**Find Your Future Path • AI Career Discovery**
MONTH 5	**AI CREATE** AI Tools	**Build with GenAI • AI Media Studio**
MONTH 6	**AI CARE** Responsible AI	**Deepfake Alert! • Your Data, Your Right**

MONTH 1

AI DISCOVER

See • Understand • Explore



Area: AI Basics

In this unit, you will explore:

Topic 1

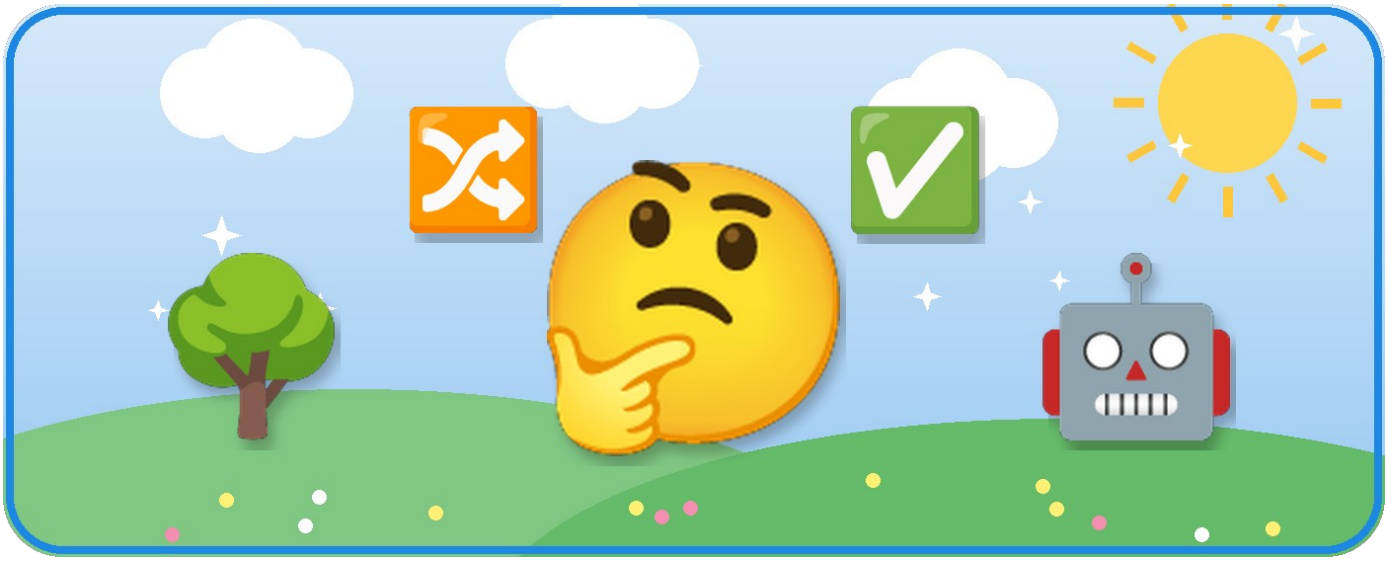
How AI Makes Choices

Topic 2

Learning from Data

TOPIC 1 • MONTH 1 • AI BASICS

How AI Makes Choices



LET'S LEARN

- AI makes choices by asking questions about the data.
- A decision tree splits choices: yes → this way, no → that way.
- AI also gives a confidence score: how sure it is!

How AI decides

Ask a question

Split yes/no

Reach a decision

Confidence %

MAGIC WORDS

Decision tree Yes/no choice map

Confidence How sure AI is

Output The final choice

💬 TECH TALK: How AI Makes Choices

💬 GAIO Squad Riya, Kabir, Bolt • online

👦 Kabir
Bolt, how does an email app decide a message is spam 📧?
10:02

🤖 Bolt
It asks questions about the data, like a flowchart. That's a decision tree 🌳.
10:04 ✓✓

👧 Riya
What questions?
10:06

🤖 Bolt
Unknown sender? Words like 'FREE prize'? Strange link 🔗? Each yes/no answer picks a branch.
10:08 ✓✓

👦 Kabir
And the last box is the answer?
10:10

🤖 Bolt
Yes, the leaf gives the output 🎯: spam or not spam.
10:12 ✓✓

👧 **Riya**

Is AI always sure?

10:14

🤖 **Bolt**

No. It gives a confidence score, like 92% spam 📊. Low confidence means a human should check.

10:16 ✓✓

👦 **Kabir**

Questions, branches, output, confidence. Got it ✅!

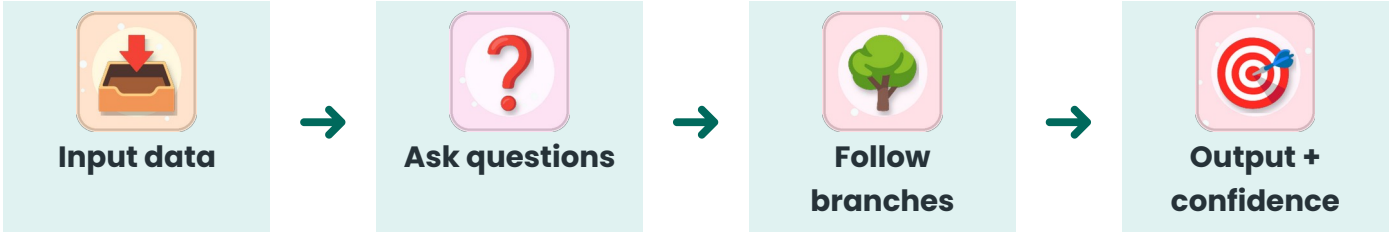
10:18

😊 Type a message...

📌 **Pinned by Bolt**

AI makes choices by asking questions about data. A decision tree follows yes/no branches to an output, often with a confidence score.

🔄 How it works



INDIA SPOTLIGHT

Indian banks and email providers use AI decision models to filter spam and flag risky transactions every second.

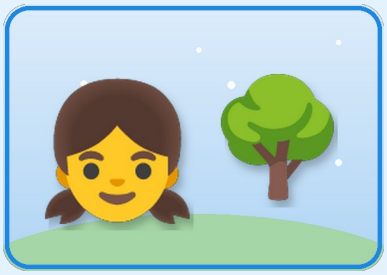
AI CHRONICLES: The Spam Detective

Riya Kabir Mr. Thomas

SCENE 1

The school inbox was flooded with fake 'You won a laptop!' emails. Teachers were missing real messages.

Riya drew a decision tree: Unknown sender? Asks for money or OTP? Suspicious link? 🔗



SCENE 2

SCENE 3

Kabir tested it on 50 old emails. It caught 42 spams but wrongly blocked a real parent email 😬.

They added a confidence score. Low-confidence emails went to a 'Check' folder for a human 👀.

SCENE 4

SCENE 5

Mr. Thomas checked the folder daily. Real emails were saved and spam dropped sharply 📉.

"AI decides fast; humans decide wisely," Riya said, as the club won the school innovation prize.

SCENE 6

🔮 What happens next? Tick ✓ one!

☐ **The club builds a phishing-SMS checker**	☐ **They block every email**	☐ **The spam writes back**

🧐 Story check: answer the questions.

1. What problem did the school inbox have?

2. What mistake did the first tree make?

3. How did the confidence score help?

Rate this story: shade the stars.

☆ ☆ ☆ ☆ ☆

LOOK AROUND YOU: Choices AI makes every day

Spam or not spam?

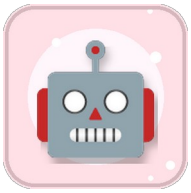
Which video to suggest?

Which route is faster?

Is this X-ray normal?

Is this payment safe?

Will it rain tomorrow?



BOLT SAYS:
I said "90% sure it's a cat". That means I could still be wrong 10% of the time!

FUN FACT!
Decision trees are one of the oldest and easiest-to-understand AI methods!

I spy AI! Draw one AI helper you saw today.

LET'S DO IT! Human Decision Tree

You need:

Animal cards

Chalk/chart

Class

Steps:

1. Draw a tree: "Can it fly?" → yes / no.
2. Add more questions: "Does it live in water?"
3. Pass each animal card down the tree.
4. Did every animal reach the right group?

THINK & TALK

Which question split the animals best?

Draw what you did!

WORKSHEET: How AI Makes Choices (Part A)

Name: _____ Date: _____

1. Match the question to the AI choice.

Is it spam? ●

Is payment safe? ●

Is X-ray normal? ●

Will it rain? ●

●

Bank AI

●

Weather AI

●

Email AI

●

Health AI

2. How a decision tree works. Write 1, 2, 3, 4.

Reach decision ☐

Get data ☐

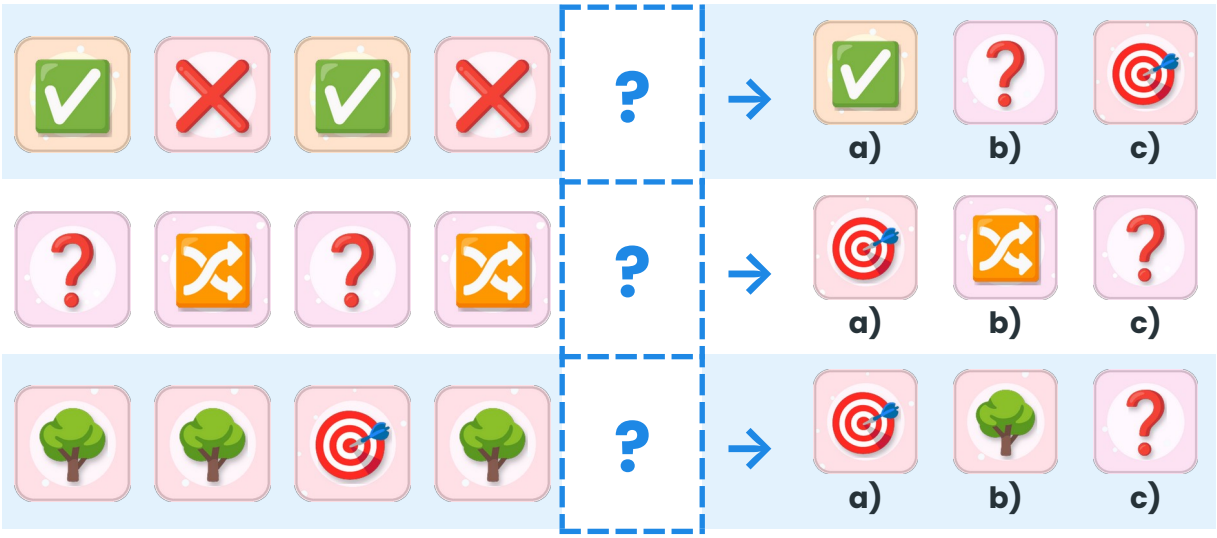
Split yes/no ☐

Ask a question ☐

WORKSHEET: How AI Makes Choices (Part B)

Name: _____ Date: _____

1. Decision path patterns. Circle what comes next.



2. Fill in the blanks. Use the word box.

WORD BOX: confidence ✿ decision tree ✿ output

A yes/no choice map is a _____.

How sure AI is, is its _____.

The final choice is the _____.

3. Bonus: Draw your favourite picture from this topic!

I drew _____

PUZZLE FUN: How AI Makes Choices

Name: _____ Date: _____

1. Odd one out! Circle it and justify your choice.

A				
B				
C				

2. Define it! Write each term in your own words with an example.

Decision tree

Meaning: _____

Example: _____

Confidence

Meaning: _____

Example: _____

Output

Meaning: _____

Example: _____

DEMO LAB: Paper decision tree + spreadsheet

Name: _____ Date: _____

Goal: Build a decision tree that classifies messages as spam or safe.

Open: chart paper / Google Sheets | **With a teacher or parent**

Steps

1. Teacher gives 12 sample messages on cards.
2. Choose 3 yes/no questions (e.g., 'Asks for OTP?').
3. Draw your tree on chart paper with branches and leaves.
4. Run all 12 messages through your tree.
5. Count correct answers and work out accuracy (correct ÷ 12 × 100).

Record your observations

My 3 questions:	
Correct answers out of 12:	
Accuracy %:	

Reflect: Which message fooled your tree? What question would fix it?

QUIZ TIME: How AI Makes Choices

Circle the right answer!

Q1. A yes/no choice map is a...

a) Decision tree

b) Apple tree

c) Road map

Q2. How sure AI is, is its...

a) Song

b) Confidence

c) Balloon

Q3. 90% sure means AI is...

a) Always right

b) Always wrong

c) Could be wrong

Q4. Decision trees split by...

a) Questions

b) Hammers

c) Pizza

Q5. The final choice is the...

a) Input

b) Output

c) Plug

MY SCORE: ____ / 5. Shade one star per correct answer.

☆ ☆ ☆ ☆ ☆

✅ TRUE OR FALSE: How AI Makes Choices

Look at the picture. Tick ✓ TRUE or FALSE.

		👍 TRUE	👎 FALSE
	A decision tree splits choices by questions.		
	90% confident means always right.		
	Email AI decides spam or not spam.		
	The final choice is the output.		
	Decision trees are a new, hard method.		
	Banks use AI to check if payments are safe.		

HOME CONNECT

Make a decision tree with your family for "What should we cook tonight?"

🔢 Self-assessment: tick one.

I know it well!

I know some.

I need practice.

EXERCISE: How AI Makes Choices

Name: _____ Date: _____

A. Tick (✓) the correct answer.

1. A yes/no flowchart AI uses to decide is a _____.

(a) decision tree ☐ **(b)** data tree ☐ **(c)** family tree ☐

2. The final answer at the end of a branch is the _____.

(a) root ☐ **(b)** output ☐ **(c)** input ☐

3. How sure AI is about an answer is its _____ score.

(a) test ☐ **(b)** speed ☐ **(c)** confidence ☐

4. Low-confidence results should be _____.

(a) checked by a human ☐ **(b)** deleted ☐ **(c)** hidden ☐

B. Fill in the blanks. Use the word box.

branch leaf questions 100

1. Each yes/no answer chooses a _____.

2. The end of a branch is a _____.

3. Spam filters ask _____ about emails.

4. Accuracy = correct ÷ total × _____.

C. Write T for True and F for False.

1. Decision trees use yes/no questions. [ ]

2. AI is always 100% confident. [ ]

3. Confidence scores help decide when humans should check. [ ]

4. A decision tree has no output. [ ]

D. Match the following. Write the letter.

Column A		Column B
1. Decision tree	____	**(a)** Final output
2. Branch	____	**(b)** Yes/no choice map
3. Leaf	____	**(c)** How sure AI is
4. Confidence	____	**(d)** Path after an answer

E. Answer in one word.

1. What map of yes/no choices does AI use? _____

2. What score shows how sure AI is? _____

3. What is the final answer called? _____

F. Answer these questions.

1. Explain how a decision tree classifies an email as spam.

2. Why are confidence scores useful?

G. Case-based questions. Read and answer.

A spam filter tree caught 42 of 50 spam emails but wrongly blocked a real parent email. The team added a 'Check' folder for low-confidence emails.

1. The wrongly blocked email was a ____ mistake.

(a) false alarm ☐ **(b)** correct result ☐ **(c)** hardware ☐

2. The 'Check' folder is for ____ results.

(a) high-confidence ☐ **(b)** low-confidence ☐ **(c)** deleted ☐

3. Who reviews the 'Check' folder?

(a) Nobody ☐ **(b)** The spammer ☐ **(c)** A human ☐

H. THINK DEEPER

Draw a 3-question decision tree to decide whether a student should carry an umbrella. What data would make it smarter?

TOPIC 2 • MONTH 1 • AI BASICS

Learning from Data

LET'S LEARN

- Supervised learning: AI learns from labelled examples.
- Unsupervised learning: AI finds groups on its own.
- Both need lots of good, fair data!

Two ways to learn

With labels

Find groups

Training

Testing

MAGIC WORDS

Supervised Learning with labels

Cluster A group of similar data

Test set Data kept to check AI

💬 TECH TALK: Learning from Data

💬 GAIO Squad Riya, Kabir, Bolt • online

👧 Riya

Bolt, how does AI learn 🧠?

10:02

🤖 Bolt

Two main ways. Supervised learning: AI learns from labelled examples, like photos tagged 'cat' 🐱 or 'dog' 🐶.

10:04 ✓✓

👦 Kabir

And the other?

10:06

🤖 Bolt

Unsupervised learning: no labels. AI finds groups (clusters) by itself 🧩.

10:08 ✓✓

👧 Riya

Like a shop grouping customers who buy similar things 🛒?

10:10

🤖 Bolt

Exactly! Then we test the AI on new data it has never seen: the test set 🧪.

10:12 ✓✓

👦 **Kabir**

Why not test on the training data?

10:14

🤖 **Bolt**

That's like giving the exam questions before the exam 📝 . The score would look great but mean nothing!

10:16 ✓✓

👧 **Riya**

Train, test, then trust ✅.

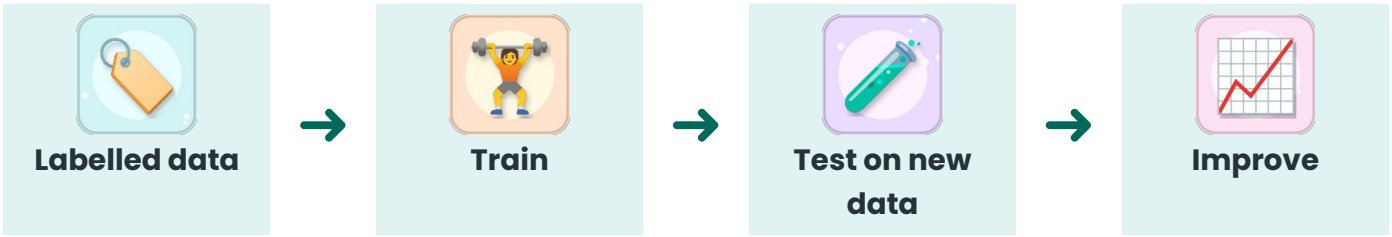
10:18

😊 Type a message...

📌 **Pinned by Bolt**

Supervised learning uses labelled examples; unsupervised learning finds clusters on its own. Always test AI on a separate test set.

🔄 How it works



INDIA SPOTLIGHT

Indian researchers train AI on labelled crop photos so farmers can identify plant diseases from a phone picture.

📚 AI CHRONICLES: The Mango Market Mystery

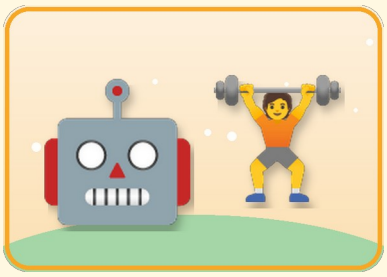
👦 Kabir | 👩 Aunt Lata (farmer) | 🤖 Bolt

SCENE 1

Aunt Lata's mango farm graded fruit by hand. It took days, and many good mangoes were wasted.

Kabir photographed 300 mangoes and labelled them 'ripe', 'raw' or 'damaged' 🏷️.

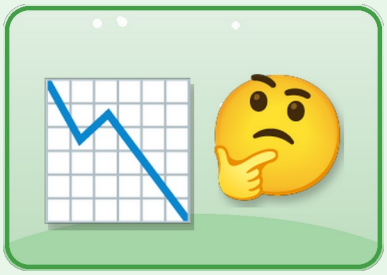
SCENE 2



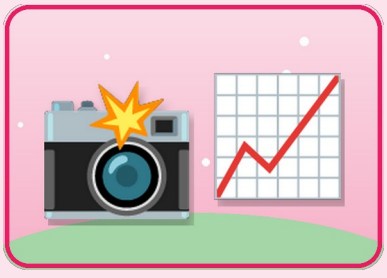
SCENE 3

Bolt trained a supervised model on 240 photos and kept 60 aside as a test set 🧪.

First test: 70%. The model had only seen mangoes in sunlight. Shade photos confused it 🌥️!



SCENE 4



SCENE 5

Kabir added photos in different light. The new score: 93% on the test set 🎉.

Later, unsupervised clustering grouped buyers by order size, so Aunt Lata could plan deliveries 🚚.

SCENE 6

🔮 What happens next? Tick ✓ one!

☐	☐	☐
Kabir builds an app for the whole village	**They stop testing the model**	**The mangoes learn to sort themselves**

🧐 Story check: answer the questions.

1. How did Kabir label the data?

2. Why did the first test score only 70%?

3. How was clustering used?

Rate this story: shade the stars.

☆ ☆ ☆ ☆ ☆

LOOK AROUND YOU: Learning in action

Labelled spam → spam filter

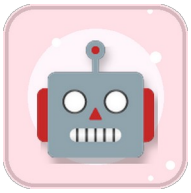
Shoppers grouped by taste

Voices → speech typing

Songs grouped by mood

Leaf photos → crop doctor

Match data → score predictor



BOLT SAYS:

With labels, I learn like a student with a teacher. Without labels, I explore like a detective!

FUN FACT!

AI teams usually keep some data aside for testing, so the AI is checked on examples it has never seen!

👀 **I spy AI! Draw one AI helper you saw today.**

LET'S DO IT! Sort Without Labels

You need:

Mixed buttons

Bowls

Teams

Steps:

1		Each team gets a pile of mixed buttons.
2		Group them any way you like: colour, size, holes...
3		Explain your groups to the class.
4		That's unsupervised learning: finding clusters!

THINK & TALK

Did different teams find different groups? Why?

Draw what you did!

WORKSHEET: Learning from Data (Part A)

Name: _____ Date: _____

1. Supervised or unsupervised? Tick ✓.

	Supervised	Unsupervised
Photos labelled cat/dog		
Grouping shoppers by habits		
Emails labelled spam		
Grouping songs by mood		

2. Fill in the blanks. Use the word box.

WORD BOX: cluster ✿ test set ✿ labels

Supervised learning uses _____.

A group of similar data is a _____.

Data kept to check AI is the _____.

WORKSHEET: Learning from Data (Part B)

Name: _____ Date: _____

1. Match the example to the learning type.

Labelled X-rays	●	●	Supervised
Grouping shoppers	●	●	Unsupervised
Labelled emails	●	●	Supervised
Grouping news topics	●	●	Unsupervised

2. Building an AI model. Write 1, 2, 3, 4.

Test on new data ☐

Collect data ☐

Use it ☐

Train ☐

PUZZLE FUN: Learning from Data

Name: _____ Date: _____

1. Odd one out! Circle it and justify your choice.

A				
B				
C				

2. Define it! Write each term in your own words with an example.

Supervised

Meaning: _____

Example: _____

Cluster

Meaning: _____

Example: _____

Test set

Meaning: _____

Example: _____

DEMO LAB: Teachable Machine (with teacher)

Name: _____ Date: _____

Goal: Train a supervised model and test it on unseen data.

Open: teachablemachine.withgoogle.com | **With a teacher or parent**

Steps

1		Create 3 labelled classes: 'Pen', 'Pencil', 'Eraser'.
2		Record 30 images per class (training data).
3		Click Train Model.
4		Test with 5 NEW objects of each kind (test set).
5		Record confidence scores and mistakes.

Record your observations

Correct on test set (out of 15):	
Lowest confidence seen:	
One mistake and why:	

Reflect: Why must the test set be different from the training data?

? QUIZ TIME: Learning from Data

Circle the right answer!

Q1. Learning with labels is...

a) Supervised

b) Unsupervised

c) Sleeping

Q2. A group of similar data is a...

a) Balloon

b) Cluster

c) Cake

Q3. The test set checks AI on...

a) Same data

b) Bad data

c) New examples

Q4. Grouping songs by mood is...

a) Unsupervised

b) Supervised

c) Singing

Q5. Both ways need...

a) Sweets

b) Good data

c) Luck

MY SCORE: _____ / 5. Shade one star per correct answer.

☆ ☆ ☆ ☆ ☆

✅ TRUE OR FALSE: Learning from Data

Look at the picture. Tick ✓ TRUE or FALSE.

		👍 TRUE	👎 FALSE
	Supervised learning uses labels.		
	Unsupervised learning finds groups on its own.		
	AI should be tested on the same data it trained on.		
	Grouping songs by mood can be unsupervised.		
	Bad data makes great AI.		
	Both learning types need good data.		

HOME CONNECT

Sort your family's photos into groups without labels. What groups did you find?

🔢 Self-assessment: tick one.

I know it well!

I know some.

I need practice.

EXERCISE: Learning from Data

Name: _____ Date: _____

A. Tick (✓) the correct answer.

1. Learning from labelled examples is ____ learning.

(a) supervised ☐ **(b)** unsupervised ☐ **(c)** unlabelled ☐

2. AI finding groups on its own is ____ learning.

(a) supervised ☐ **(b)** unsupervised ☐ **(c)** copied ☐

3. Data kept aside to check AI is the ____.

(a) training set ☐ **(b)** label set ☐ **(c)** test set ☐

4. A group found by unsupervised learning is a ____.

(a) cluster ☐ **(b)** label ☐ **(c)** loop ☐

B. Fill in the blanks. Use the word box.

labelled **misleading** **customers** **data**

1. Photos tagged 'cat' are ____ data.

2. Testing on training data gives a ____ score.

3. Shops cluster ____ with similar buying habits.

4. Good AI needs good, fair ____.

C. Write T for True and F for False.

1. Supervised learning uses labels. [ ]

2. The test set should be the same as the training data. [ ]

3. Unsupervised learning finds clusters. [ ]

4. More varied data can improve accuracy. [ ]

D. Match the following. Write the letter.

Column A		Column B
1. Supervised	____	**(a)** Unseen data for checking
2. Unsupervised	____	**(b)** Learns from labels
3. Test set	____	**(c)** Group of similar items
4. Cluster	____	**(d)** Finds groups itself

E. Answer in one word.

1. Which learning uses labels? _____

2. What is a group of similar data called? _____

3. What data checks AI fairly? _____

F. Answer these questions.

1. Differentiate supervised and unsupervised learning with examples.

2. Why should AI be tested on new data?

G. Case-based questions. Read and answer.

A mango-sorting model trained only on sunlit photos scored 70% on the test set. After adding photos in shade and different light, it scored 93%.

1. The first model lacked ____ data.

(a) varied ☐ **(b)** labelled ☐ **(c)** any ☐

2. The mango model used ____ learning.

(a) unsupervised ☐ **(b)** supervised ☐ **(c)** no ☐

3. The score rose because ____.

(a) the test was easier ☐ **(b)** mangoes changed ☐ **(c)** the data improved ☐

H. THINK DEEPER

You want an AI to group your school's library books without labels. What features would you use, and what groups might it find?

UNIT REVIEW: AI DISCOVER

Name: _____ Date: _____

Circle the right answer!

Q1. A yes/no choice map is a...

a) Decision tree

b) Road map

c) Apple tree

Q2. Learning with labels is...

a) Unsupervised

b) Supervised

c) Resting

Q3. A group of similar data is a...

a) Balloon

b) Cake

c) Cluster

Q4. How sure AI is, is its...

a) Confidence

b) Song

c) Toy

Q5. Data kept to check AI is the...

a) Bin

b) Test set

c) Gift

Q6. The final choice is the...

a) Input

b) Plug

c) Output

WORD SEARCH: AI DISCOVER

🔍 **Find these words. They go across → or down ↓.**

CONFIDENCE

OUTPUT

SUPERVISED

CLUSTER

TEST SET

LABELS

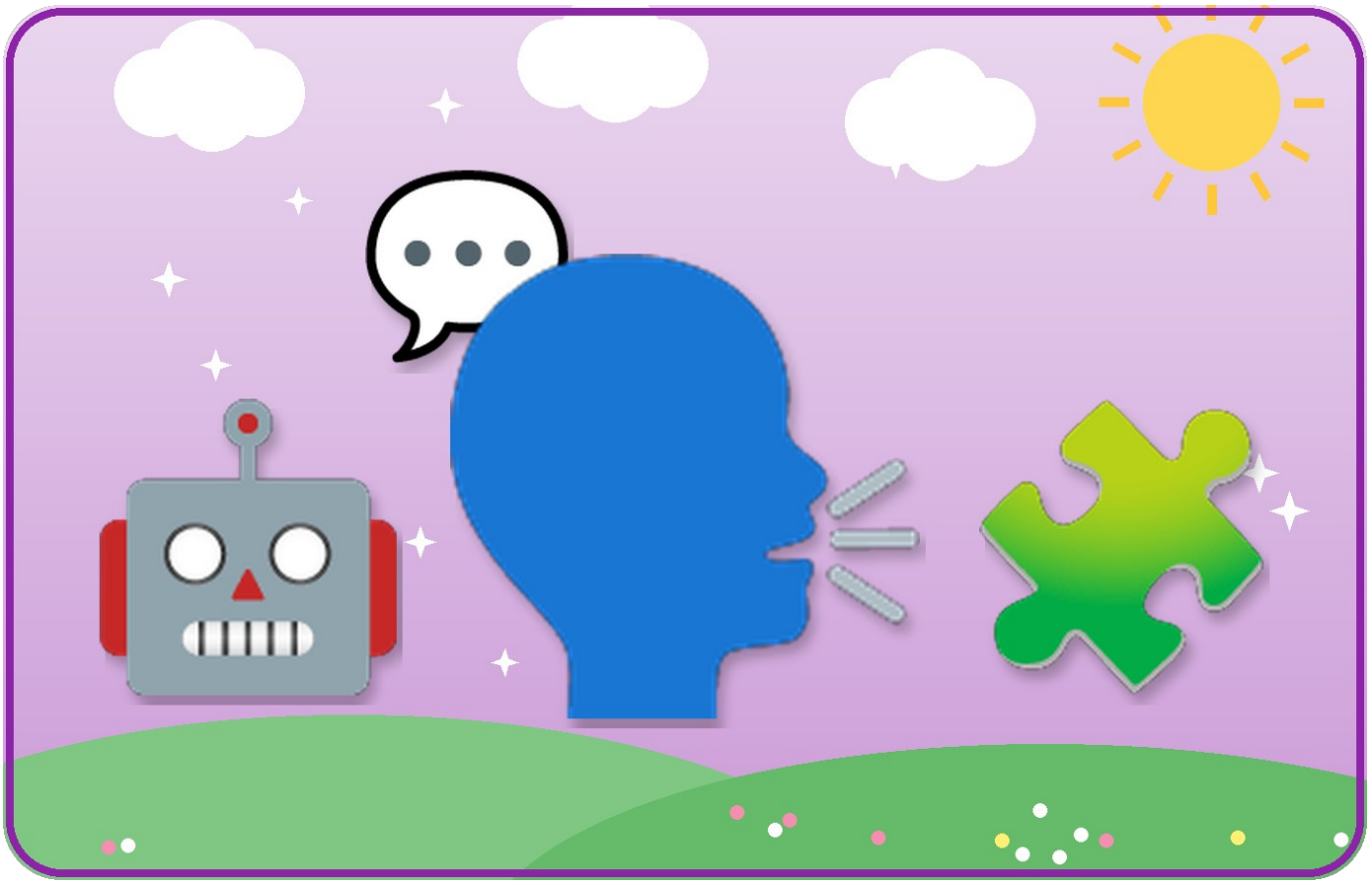
H	N	O	L	A	B	E	L	S	F
C	O	N	F	I	D	E	N	C	E
M	T	M	C	P	S	D	F	S	N
T	E	S	T	S	E	T	M	R	A
R	B	K	W	U	U	N	F	F	S
H	A	G	T	T	H	N	S	M	U
M	P	I	C	L	U	S	T	E	R
S	U	P	E	R	V	I	S	E	D
O	U	T	P	U	T	T	W	A	N
S	E	S	T	G	O	B	R	M	U

I finished AI DISCOVER! Teacher's sign: _____

MONTH 2

AI CONNECT

Think • Instruct • Code



Area: Communicate with AI

In this unit, you will explore:

Topic 3

Python Playground

Topic 4

Code a Solution

TOPIC 3 • MONTH 2 • COMMUNICATE WITH AI

Python Playground

LET'S LEARN

- Lists store many items: fruits = ["apple", "mango"]
- A for loop visits each item in a list.
- Functions are reusable blocks: def greet():

Python toys

list [ ]

for loop

def function

len()

MAGIC WORDS

List Many items in one

Function Reusable code block

Iterate Go through each item

💬 TECH TALK: Python Playground

💬 GAIO Squad Riya, Kabir, Bolt • online

👦 **Kabir**
Bolt, I have 30 friends' names. 30 variables 😩?
10:02

🤖 **Bolt**
No! Use a list 📋: friends = ['Riya', 'Kabir', 'Aman']
10:04 ✓✓

👧 **Riya**
How do we print each one?
10:06

🤖 **Bolt**
Iterate with a for loop 🔁: for f in friends: print('Hi', f)
10:08 ✓✓

👦 **Kabir**
Can we add a name?
10:10

🤖 **Bolt**
friends.append('Meenu') ➕. And len(friends) tells how many.
10:12 ✓✓

👧 **Riya**
What if we greet people many times?
10:14

Bolt

Make a function 🧰: def greet(name): print('Welcome', name). Then call greet('Riya').

10:16 ✓✓

Kabir

Lists store, loops repeat, functions reuse 🚀!

10:18

😊 Type a message...

📌 **Pinned by Bolt**

Lists store many items. A for loop iterates over each item. Functions (def) are reusable blocks of code.

🔄 How it works

INDIA SPOTLIGHT

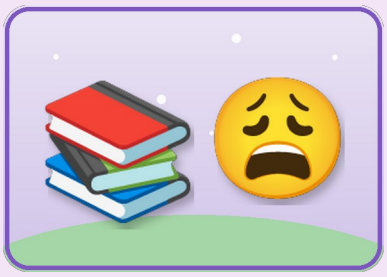
Many Indian students learn Python through free online courses and national platforms that teach coding in regional languages too.

AI CHRONICLES: The Library List

Riya

Librarian Ms. Jaya

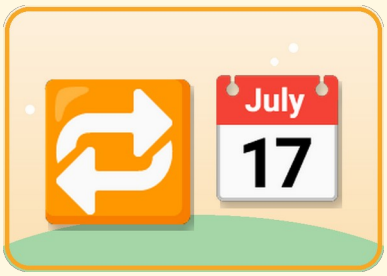
Python


SCENE 1

Ms. Jaya tracked 500 borrowed books in a messy notebook. Overdue books were often forgotten.

Riya stored the borrowers' names in a Python list: borrowers = ['Aman', 'Meenu', ...]

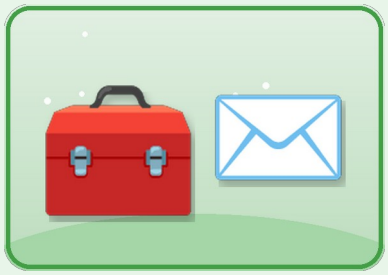
SCENE 2



SCENE 3

A for loop checked each record and printed names whose books were overdue ⏰.

She wrote a function, remind(name), to create a polite reminder for each student.


SCENE 4

SCENE 5

A bug! One name repeated 50 times. She had put the print inside the wrong loop. Fixed ✅!

Overdue books dropped by half. Ms. Jaya smiled: "Lists, loops and kindness!"

SCENE 6

🔮 What happens next? Tick ✓ one!

☐	☐	☐
Riya adds a book-recommendation feature	**Riya deletes the list**	**The books start returning themselves**

🧐 Story check: answer the questions.

1. What did Riya store in a list?

2. What did the function do?

3. What caused the bug?

Rate this story: shade the stars.

☆ ☆ ☆ ☆ ☆

LOOK AROUND YOU: Where lists & loops help

Shopping list app

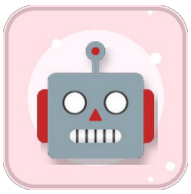
Team player list

Library books

Music playlist

Marks of a class

Game high scores



BOLT SAYS:

Try: for f in fruits: print(f) — I'll print every fruit, one by one!

FUN FACT!

A function is like a recipe. Write it once, use it many times!

👀 **I spy AI! Draw one AI helper you saw today.**

LET'S DO IT! Human List & Loop

You need:

Number cards

5 students

A caller

Steps:

1		5 students stand in a line: they are a list.
2		Each holds an index card: 0, 1, 2, 3, 4.
3		Caller runs a loop: each student says their name.
4		Make a function: wave() — everyone waves when called!

THINK & TALK

Why does Python count from 0, not 1?

Draw what you did!

WORKSHEET: Python Playground (Part A)

Name: _____ Date: _____

1. What will Python print? Use the word box.

WORD BOX: 3 ✿ mango ✿ apple

fruits = ["apple","mango","kiwi"]; print(fruits[0]) → ____

print(len(fruits)) → ____

print(fruits[1]) → ____

2. Match the Python word to its meaning.

[ ] ●	● Counts items
for ●	● Makes a function
def ●	● A list
len() ●	● Loops

WORKSHEET: Python Playground (Part B)

Name: _____ Date: _____

1. Circle the correct Python code.

nums = [1, 2, 3]

nums = (1 2 3

for n in nums:

for n nums

def hello():

def hello

2. Complete the code. Use the word box.

WORD BOX: for ✿ def ✿ len

____ c in colours: print(c)

____ greet(): print("Hi")

print(____(colours))

PUZZLE FUN: Python Playground

Name: _____ Date: _____

1. Odd one out! Circle it and justify your choice.

2. Define it! Write each term in your own words with an example.

List

Meaning: _____

Example: _____

Function

Meaning: _____

Example: _____

Iterate

Meaning: _____

Example: _____

🧪 DEMO LAB: Online Python editor

Name: _____ Date: _____

🎯 **Goal:** Use lists, loops and functions.

🔗 **Open:** any browser Python editor / Google Colab | 👩‍🏫 **With a teacher or parent**

👣 Steps

1 📋 fruits = ['apple', 'mango', 'banana']

2 ➕ fruits.append('guava'); print(len(fruits))

3 🔁 for f in fruits: print('I like', f)

4 🧰 def greet(name): print('Welcome,', name)

5 📞 Call greet() for 3 classmates using a loop.

🔬 Record your observations

Output of len(fruits):	
Output of your loop (first line):	
Your function's output:	

💭 **Reflect:** How do functions save time when your program grows?

? QUIZ TIME: Python Playground

Circle the right answer!

Q1. Many items in one variable is a...

a) List

b) Balloon

c) Cake

Q2. A for loop goes through...

a) Trees

b) Each item

c) Nothing

Q3. def is used to make a...

a) Cake

b) Song

c) Function

Q4. Python list index starts at...

a) 0

b) 1

c) 10

Q5. len([1,2,3,4]) is...

a) 3

b) 4

c) 5

MY SCORE: _____ / 5. Shade one star per correct answer.

✅ TRUE OR FALSE: Python Playground

Look at the picture. Tick ✓ TRUE or FALSE.

	👍 TRUE	👎 FALSE
A list stores many items.		
Python lists start at index 0.		
def makes a loop.		
len() counts items.		
A for loop visits each item.		
fruits[1] is the first item.		

HOME CONNECT

Write your family members as a Python list on paper. How many items does len() give?

🔢 Self-assessment: tick one.

I know it well!

I know some.

I need practice.

EXERCISE: Python Playground

Name: _____ Date: _____

A. Tick (✓) the correct answer.

1. A Python structure that stores many items is a ____.

(a) list ☐ **(b)** loop ☐ **(c)** label ☐

2. To go through each item in a list, use a ____ loop.

(a) while-not ☐ **(b)** for ☐ **(c)** if ☐

3. Reusable blocks of code are ____.

(a) comments ☐ **(b)** errors ☐ **(c)** functions ☐

4. To add an item to a list, use ____.

(a) append() ☐ **(b)** print() ☐ **(c)** def ☐

B. Fill in the blanks. Use the word box.

def **items** **iterate** **brackets**

1. Functions start with the keyword ____.

2. len(list) gives the number of ____.

3. Going through items one by one is to ____.

4. Lists use square ____.

C. Write T for True and F for False.

1. A list can store many names. [ ]

2. A function can only be used once. [ ]

3. for f in fruits visits each fruit. [ ]

4. append() deletes items. [ ]

D. Match the following. Write the letter.

Column A		Column B
1. List	____	**(a)** Creates a function
2. for loop	____	**(b)** Stores many items
3. def	____	**(c)** Counts items
4. len()	____	**(d)** Repeats for each item

E. Answer in one word.

1. Which keyword creates a function? _____

2. Which function counts list items? _____

3. Which method adds to a list? _____

F. Answer these questions.

1. Write Python code to store 3 city names in a list and print each.

2. Why are functions useful?

G. Case-based questions. Read and answer.

Riya's library program stores borrowers in a list, loops through records and uses remind(name) to print reminders. A misplaced print made one name repeat 50 times.

1. The borrowers are stored in a ____.

(a) list ☐ **(b)** function ☐ **(c)** comment ☐

2. remind(name) is a ____.

(a) loop ☐ **(b)** function ☐ **(c)** list ☐

3. The repeating name was a ____.

(a) feature ☐ **(b)** prize ☐ **(c)** bug ☐

H. THINK DEEPER

Write a function that takes a list of marks and returns the highest mark. Explain each line.

TOPIC 4 • MONTH 2 • COMMUNICATE WITH AI

Code a Solution

LET'S LEARN

- Coders solve real problems step by step.
- Understand → break into parts → code → test → improve.
- Debugging is part of the fun!

Problem-solving steps

Understand

Break down

Code

Debug

MAGIC WORDS

Decompose Break into parts

Debug Find & fix errors

Error A problem in code

💬 TECH TALK: Code a Solution

💬 GAIO Squad Riya, Kabir, Bolt • online

👧 Riya
Our class wastes time calculating test averages by hand 🧮.
10:02

🤖 Bolt
Let's code a solution! Step 1: understand the problem 🔍.
10:04 ✓✓

👦 Kabir
Input marks, output average and grade.
10:06

🤖 Bolt
Step 2: decompose 🧩. Get marks find average decide grade show result.
10:08 ✓✓

👧 Riya
Average = sum(marks) / len(marks)!
10:10

🤖 Bolt
Right. Then if avg >= 90: grade 'A' 🏅, elif avg >= 75: 'B', else 'C'.
10:12 ✓✓

👦 **Kabir**

I typed it and got an error 🐞!

10:14

🤖 **Bolt**

Read the message. A missing colon after if? Debugging is part of coding 🔧.

10:16 ✓✓

👧 **Riya**

Understand, decompose, code, test, improve ✅!

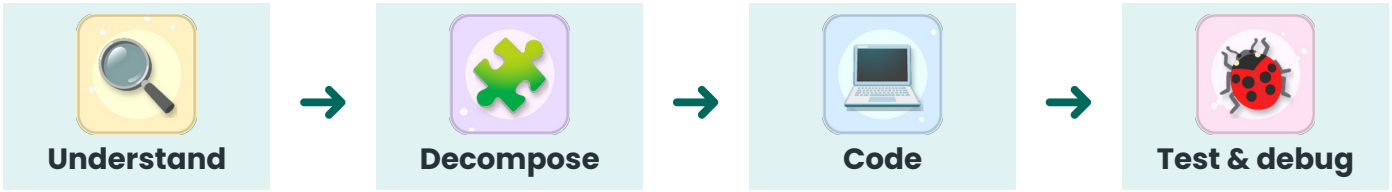
10:18

😊 Type a message...

📌 **Pinned by Bolt**

Coders solve problems step by step: understand, decompose, code, test and improve. Errors are clues; debugging fixes them.

🔄 How it works



INDIA SPOTLIGHT

Smart India Hackathon invites students to code solutions for real problems given by ministries and industries.

AI CHRONICLES: Hackathon in 24 Hours

Kabir Riya Judge

SCENE 1

At the school hackathon, the challenge was: 'Reduce lunch-queue waiting time' 🍱.

The team first interviewed students. The real problem: everyone came at the same time.

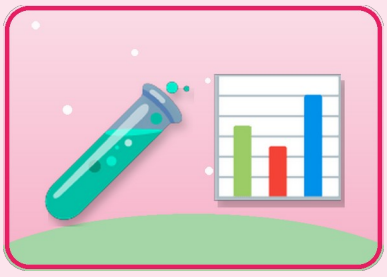
SCENE 2

SCENE 3

They decomposed it: collect class timings assign slots show the slot collect feedback.

At 3 a.m., their code crashed: a TypeError! They read the message and fixed a text-number mix-up 🔧.

SCENE 4



SCENE 5

Tests with 3 classes cut waiting from 20 to 6 minutes 📉.

The judge said, "You understood the problem before coding. That's why it worked!"

SCENE 6

🔮 What happens next? Tick ✓ one!

☐	☐	☐
The school uses their slot system daily	**They code without understanding next time**	**The lunch queue writes code**

🧐 Story check: answer the questions.

1. What was the real problem?

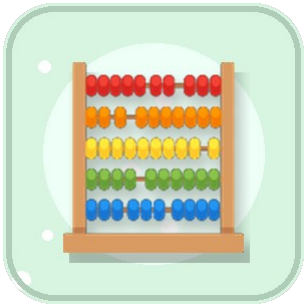
2. What were the decomposed steps?

3. Why did the judge praise the team?

Rate this story: shade the stars.

☆ ☆ ☆ ☆ ☆

LOOK AROUND YOU: Student-sized coding projects



Average marks calculator

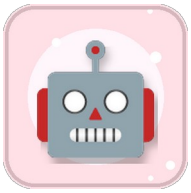
Water-use tracker

Quiz game

Fever checker

Password strength checker

Homework reminder



BOLT SAYS:
An error message is not a failure. It's a clue to fix your code!

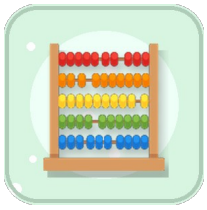
FUN FACT!
The word "bug" became famous when a real moth was found stuck in an early computer in 1947!

I spy AI! Draw one AI helper you saw today.

LET'S DO IT! Average Marks Calculator

You need:

Paper code



Calculator

A partner

Steps:

1

marks = [80, 70, 90]

2

total = sum(marks)

3

average = total / len(marks)

4

Check with a calculator. Is it 80?

THINK & TALK

What would happen if the marks list was empty?

Draw what you did!

WORKSHEET: Code a Solution (Part A)

Name: _____ Date: _____

1. Solving a problem with code. Write 1, 2, 3, 4.

Debug & improve ☐

Understand the problem ☐

Write code ☐

Break it down ☐

2. Find the bug! Correct or wrong code? Tick ✓.

		Correct	Has a bug
	print("Hello")		
	print("Hello"		
	nums = [1, 2, 3]		
	if x > 5 print(x)		
	for n in nums:		

WORKSHEET: Code a Solution (Part B)

Name: _____ Date: _____

1. Match the project to the code idea.

Average marks ●	● if temp > 100
Quiz game ●	● for loop 1 to 10
Fever checker ●	● sum / len
Times tables ●	● input + if

2. Debug it! Use the word box.

WORD BOX: : ✿ " ✿)

print("Hi" ← missing ____

if x > 5 ← missing ____

print("Hello) ← missing ____

PUZZLE FUN: Code a Solution

Name: _____ Date: _____

1. Odd one out! Circle it and justify your choice.

A				
B				
C				

2. Define it! Write each term in your own words with an example.

Decompose

Meaning: _____

Example: _____

Debug

Meaning: _____

Example: _____

Error

Meaning: _____

Example: _____

🧪 DEMO LAB: Online Python editor

Name: _____ Date: _____

🎯 **Goal:** Code a marks-average and grade calculator, then debug it.

🔗 **Open:** any browser Python editor / Google Colab | 👩‍🏫 **With a teacher or parent**

👣 Steps

1		marks = [78, 92, 85, 64, 90]
2		avg = sum(marks) / len(marks); print(avg)
3		if avg >= 90: print('A') / elif avg >= 75: print('B') / else: print('C')
4		Remove a colon on purpose. Read the error message, then fix it.
5		Test with 3 different mark lists.

🔬 Record your observations

Average for the first list:	
Grade shown:	
Error message you saw:	

💭 **Reflect:** Why is reading error messages an important coding skill?

? QUIZ TIME: Code a Solution

Circle the right answer!

Q1. Breaking a problem into parts is...

a) Decompose

b) Explode

c) Sleep

Q2. Finding & fixing errors is...

a) Singing

b) Debugging

c) Sweeping

Q3. An error message is a...

a) Failure

b) Toy

c) Clue

Q4. sum([2,3,5]) is...

a) 10

b) 9

c) 5

Q5. The famous first bug was a...

a) Ant

b) Moth

c) Bee

MY SCORE: _____ / 5. Shade one star per correct answer.

☆ ☆ ☆ ☆ ☆

✅ TRUE OR FALSE: Code a Solution

Look at the picture. Tick ✓ TRUE or FALSE.

		👍 TRUE	👎 FALSE
	Decompose means break into parts.		
	Error messages give clues.		
	Debugging is a sign of failure.		
	average = total / count.		
	A moth was a famous early bug.		
	Never test your code.		

HOME CONNECT

Find a small problem at home. Write the steps a program could use to solve it.

🔢 Self-assessment: tick one.

I know it well!

I know some.

I need practice.

EXERCISE: Code a Solution

Name: _____ Date: _____

A. Tick (✓) the correct answer.

1. Breaking a problem into smaller parts is to ____.

(a) decompose ☐ **(b)** delete ☐ **(c)** decorate ☐

2. Finding and fixing mistakes in code is ____.

(a) designing ☐ **(b)** debugging ☐ **(c)** downloading ☐

3. The first step in solving a problem is to ____ it.

(a) code ☐ **(b)** submit ☐ **(c)** understand ☐

4. A missing colon after 'if' causes an ____.

(a) error ☐ **(b)** average ☐ **(c)** upgrade ☐

B. Fill in the blanks. Use the word box.

len if clues improve

1. Average = sum(marks) / ____(marks).

2. elif means 'else ____'.

3. Error messages are ____ to the bug.

4. After testing, we ____ the code.

C. Write T for True and F for False.

1. Debugging is a normal part of coding. [ ]

2. Good coders never get errors. [ ]

3. Testing with different inputs is useful. [ ]

4. You should code before understanding the problem. [ ]

D. Match the following. Write the letter.

Column A		Column B
1. Understand	____	**(a)** Fix errors
2. Decompose	____	**(b)** Know the real problem
3. Debug	____	**(c)** Try different inputs
4. Test	____	**(d)** Break into parts

E. Answer in one word.

1. What is fixing code errors called? _____

2. What is breaking a problem into parts? _____

3. Which keyword means 'else if'? _____

F. Answer these questions.

1. List the five steps of solving a problem with code.

2. Why should we test code with different inputs?

G. Case-based questions. Read and answer.

A hackathon team interviewed students and found lunch queues were long because everyone arrived together. They coded a slot system; a TypeError at 3 a.m. was fixed by reading the error.

1. Interviewing students helped them ____ the problem.

(a) understand ☐ **(b)** hide ☐ **(c)** skip ☐

2. The TypeError was fixed by ____.

(a) giving up ☐ **(b)** reading the error message ☐ **(c)** restarting the PC ☐

3. The solution was a ____ system.

(a) music ☐ **(b)** grading ☐ **(c)** time-slot ☐

H. THINK DEEPER

Pick a problem in your school. Decompose it into steps and write the inputs, process and outputs of a program to solve it.

UNIT REVIEW: AI CONNECT

Name: _____ Date: _____

Circle the right answer!

Q1. Many items in one is a...

a) List

b) Balloon

c) Pizza

Q2. def creates a...

a) Cake

b) Function

c) Song

Q3. Finding & fixing errors is...

a) Singing

b) Sweeping

c) Debugging

Q4. List index starts at...

a) 0

b) 1

c) 10

Q5. Break into parts is...

a) Explode

b) Decompose

c) Sleep

Q6. len([5,6]) is...

a) 5

b) 6

c) 2

WORD SEARCH: AI CONNECT

🔍 **Find these words. They go across → or down ↓.**

LIST

FUNCTION

ITERATE

DECOMPOSE

DEBUG

ERROR

M	D	P	F	N	P	S	H	R	I
R	E	S	S	M	P	P	M	U	T
P	C	F	U	N	C	T	I	O	N
R	O	H	L	F	W	I	I	R	K
K	M	S	T	S	D	T	S	W	U
O	P	K	G	R	E	E	S	E	M
W	O	C	L	A	B	R	G	R	D
B	S	U	B	I	U	A	U	R	H
D	E	S	E	I	G	T	H	O	G
B	O	L	I	S	T	E	B	R	B

I finished AI CONNECT! Teacher's sign: _____

MONTH 3

AI SOLVE

Problems • Solutions • Impact

Area: AI Applications

In this unit, you will explore:

Topic 5

AI for Better Health

Topic 6

AI for Better Cities

TOPIC 5 • MONTH 3 • AI APPLICATIONS

AI for Better Health

LET'S LEARN

- AI helps doctors detect diseases early.
- It reads scans, predicts risks and supports rural health.
- Doctors decide. AI assists. Patients' data must stay private.

Health AI tools

Scan reader

Health apps

Medicine research

Emergency routing

MAGIC WORDS

Diagnosis Finding the illness

Telemedicine Doctor over phone/video

Confidential Kept private

💬 TECH TALK: AI for Better Health

💬 GAIO Squad Riya, Kabir, Bolt • online

👦 Kabir

My grandpa's village has no eye specialist 👁️. Can AI help?

10:02

🤖 Bolt

Yes! AI can scan eye photos to detect diabetic eye disease early 📷.

10:04 ✓✓

👧 Riya

So AI replaces doctors?

10:06

🤖 Bolt

No. AI assists, doctors decide 👩‍⚕️. AI flags risks; doctors confirm and treat.

10:08 ✓✓

👦 Kabir

What else?

10:10

🤖 Bolt

Reading X-rays 🩻, predicting outbreaks and telemedicine: video consultations from far away 📱.

10:12 ✓✓

Riya
Health data is private, right?
10:14

Bolt
Very! Medical records are confidential 🔒. Only the patient and their doctors should see them.
10:16 ✓✓

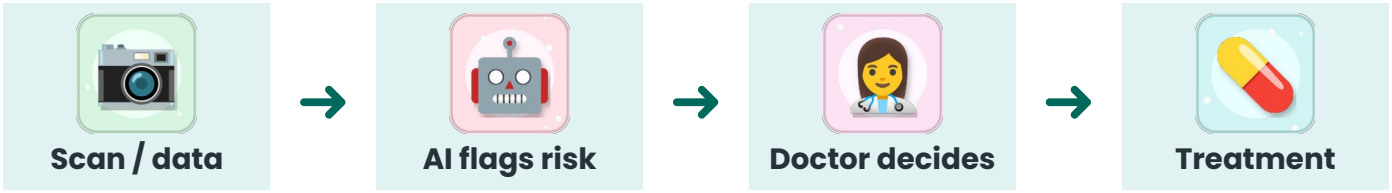
Kabir
Early detection, expert decisions, private data ❤️.
10:18

Type a message...

📌 **Pinned by Bolt**

AI helps detect diseases early, reads scans and supports telemedicine. Doctors make the final decision, and patient data must stay confidential.

🔄 How it works

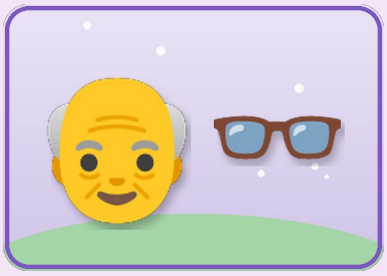


INDIA SPOTLIGHT

eSanjeevani, India's national telemedicine service, has connected millions of patients with doctors through online consultations.

AI CHRONICLES: The Eye Camp

Riya Grandpa Dr. Meera

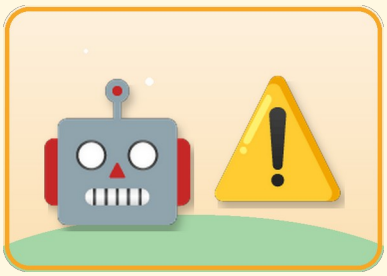


SCENE 1

Grandpa's vision was getting blurry, but the nearest eye specialist was 80 km away 🚌.

A health camp came to the village. A nurse photographed his eyes with a special camera.

SCENE 2

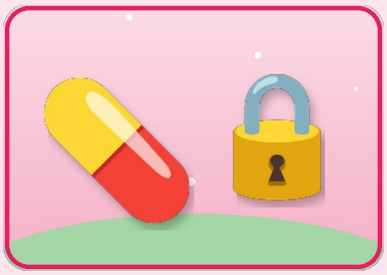


SCENE 3

AI analysed the photo in minutes and flagged a possible diabetic eye problem.

Through telemedicine, Dr. Meera reviewed the scan from the city and confirmed it ✅.

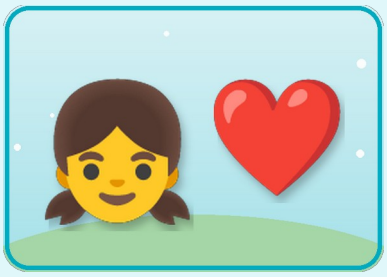
SCENE 4



SCENE 5

Grandpa got early treatment. His records were stored securely, shared only with his doctors.

Riya wrote in her diary: "AI found it early. The doctor decided. Grandpa can see!"



SCENE 6

🔮 What happens next? Tick ✓ one!

☐	☐	☐
Riya volunteers at the next health camp	**Grandpa ignores the treatment**	**The camera becomes a doctor**

🧐 Story check: answer the questions.

1. Why couldn't Grandpa see a specialist easily?

2. What did AI do, and what did the doctor do?

3. How was his data protected?

Rate this story: shade the stars.

☆ ☆ ☆ ☆ ☆

LOOK AROUND YOU: AI improving health in India

Eye scans for diabetes

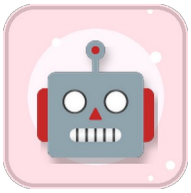
TB detection in X-rays

Telemedicine for villages

Fitness & heart tracking

Faster medicine discovery

Health info in local languages



BOLT SAYS:

I can scan thousands of images, but a doctor understands the whole person!

FUN FACT!

AI tools in India help screen eyes for diabetes damage, even in small clinics!

I spy AI! Draw one AI helper you saw today.

LET'S DO IT! Health Data Detective

You need:

Step data cards

Graph paper

Pencil

Steps:

1

Teacher gives a week of (pretend) step counts.

2

Plot them on a graph.

3

Find the pattern: which days were active?

4

Suggest one healthy habit based on the data.

THINK & TALK

Why must health data be kept confidential?

Draw what you did!

WORKSHEET: AI for Better Health (Part A)

Name: _____ Date: _____

1. Match the health AI to its job.

Eye scanner ●

Telemedicine ●

Smart watch ●

X-ray AI ●

●

Tracks heart rate

●

Spots TB signs

●

Checks diabetic eyes

●

Helps villages

2. Fill in the blanks. Use the word box.

WORD BOX: diagnosis ✿ confidential ✿ telemedicine

Finding the illness is _____.

A doctor on video call is _____.

Health data must be _____.

WORKSHEET: AI for Better Health (Part B)

Name: _____ Date: _____

1. Can AI do this alone, or must a doctor? Tick ✓.

	AI helps	Doctor decides
Scan 1000 X-rays fast		
Choose final treatment		
Medicine reminders		
Comfort a patient		

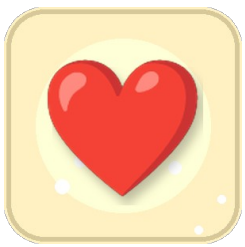
2. Circle the health data that must be kept private.

Medical reports

Fav colour

Medicines taken

Fav sport


Heart rate

Blood group

PUZZLE FUN: AI for Better Health

Name: _____ Date: _____

1. Odd one out! Circle it and justify your choice.

A				
B				
C				

2. Define it! Write each term in your own words with an example.

Diagnosis

Meaning: _____

Example: _____

Telemedicine

Meaning: _____

Example: _____

Confidential

Meaning: _____

Example: _____

🧪 DEMO LAB: Health-AI case cards (unplugged)

Name: _____ Date: _____

🎯 **Goal:** Explore how AI assists, and where humans must decide.

🔗 **Open:** printed case cards from teacher | 👩‍🏫 **With a teacher or parent**

👣 Steps

1. Read 4 cases: eye scan, X-ray, fitness band alert, symptom-checker chatbot.
2. For each, write what AI does.
3. Write what the doctor must still decide.
4. Circle the private data in each case.
5. Rate each case: helpful / risky / needs care.

🔬 Record your observations

Most helpful use:	
Riskiest use:	
Private data found:	

💭 **Reflect:** Why should a doctor, not AI, make the final health decision?

QUIZ TIME: AI for Better Health

Circle the right answer!

Q1. Who makes the final health decision?

a) Doctor

b) Robot

c) App

Q2. Doctor over video call is...

a) Gaming

b) Telemedicine

c) TV

Q3. Health data should be...

a) Public

b) Shared online

c) Confidential

Q4. AI can help detect...

a) TB in X-rays

b) Songs

c) Rainbows

Q5. Finding the illness is...

a) Cooking

b) Diagnosis

c) Painting

MY SCORE: _____ / 5. Shade one star per correct answer.

✅ TRUE OR FALSE: AI for Better Health

Look at the picture. Tick ✓ TRUE or FALSE.

	👍 TRUE	👎 FALSE
AI can screen eyes for diabetes damage.		
AI replaces doctors completely.		
Telemedicine helps village patients.		
Health data should be confidential.		
Post medical reports online.		
AI can help find TB in X-rays.		

HOME CONNECT

Track your family's daily steps for a week (with a phone or by counting). Make a graph!

🔢 Self-assessment: tick one.

I know it well!

I know some.

I need practice.

EXERCISE: AI for Better Health

Name: _____ Date: _____

A. Tick (✓) the correct answer.

1. AI can help detect diseases ____.

(a) early ☐ **(b)** never ☐ **(c)** only after surgery ☐

2. The final health decision is made by the ____.

(a) app ☐ **(b)** doctor ☐ **(c)** camera ☐

3. Video consultation with a distant doctor is ____.

(a) teleshopping ☐ **(b)** telescope ☐ **(c)** telemedicine ☐

4. Medical records must be kept ____.

(a) confidential ☐ **(b)** public ☐ **(c)** on posters ☐

B. Fill in the blanks. Use the word box.

scans **decide** **eSanjeevani** **diagnosis**

1. AI can read ____ like X-rays.

2. AI assists; doctors ____.

3. India's telemedicine service is ____.

4. Finding a disease is called ____.

C. Write T for True and F for False.

1. AI can analyse eye photos. [ ]

2. AI should replace all doctors. [ ]

3. Telemedicine helps rural patients. [ ]

4. Health data can be shared with anyone. [ ]

D. Match the following. Write the letter.

Column A		Column B
1. Diagnosis	____	**(a)** Kept private
2. Telemedicine	____	**(b)** Identifying a disease
3. Confidential	____	**(c)** Image of bones
4. X-ray	____	**(d)** Remote consultation

E. Answer in one word.

1. Who makes the final treatment decision? _____

2. What is remote consultation called? _____

3. Should health data be private? _____

F. Answer these questions.

1. Give three ways AI helps healthcare.

2. Why must patient data be confidential?

G. Case-based questions. Read and answer.

At a village camp, AI flagged a possible eye problem from Grandpa's photo. Dr. Meera confirmed it through telemedicine, and his records were shared only with his doctors.

1. AI's role was to _____.

(a) flag a risk ☐ **(b)** do surgery ☐ **(c)** give medicine ☐

2. Dr. Meera used _____.

(a) a radio ☐ **(b)** telemedicine ☐ **(c)** a newspaper ☐

3. Grandpa's data was _____.

(a) posted online ☐ **(b)** sold ☐ **(c)** kept confidential ☐

H. THINK DEEPER

If an AI health app gives a wrong result, who should be responsible: the app makers, the doctor or the user? Justify your answer.

TOPIC 6 • MONTH 3 • AI APPLICATIONS

AI for Better Cities

LET'S LEARN

- Smart cities use sensors, data and AI to work better.
- They manage traffic, water, power, waste and safety.
- Good smart cities are planned for ALL citizens!

Smart city parts

Smart traffic

Smart lights

Water monitoring

Smart bins

MAGIC WORDS

Sensor Collects city data

Smart city City using tech well

Accessible Usable by everyone

💬 TECH TALK: AI for Better Cities

💬 GAIO Squad Riya, Kabir, Bolt • online

👧 Riya
Bolt, why do traffic signals near my school stay red even when roads are empty 🚦?
10:02

🤖 Bolt
They're on fixed timers. Smart signals use sensors and cameras to adjust to real traffic 📷.
10:04 ✓✓

👦 Kabir
What's a smart city?
10:06

🤖 Bolt
A city that uses sensors, data and AI to manage traffic, water 💧, power ⚡ and waste.
10:08 ✓✓

👧 Riya
Like bins that say when they're full 🗑️?
10:10

🤖 Bolt
Yes! Trucks only go where needed, saving fuel.
10:12 ✓✓

👦 **Kabir**

Is a smart city for everyone?

10:14

🤖 **Bolt**

It should be! Accessible design means ramps ♿, audio signals and apps in local languages.

10:16 ✓✓

👧 **Riya**

Smart means fair for all 🏙️!

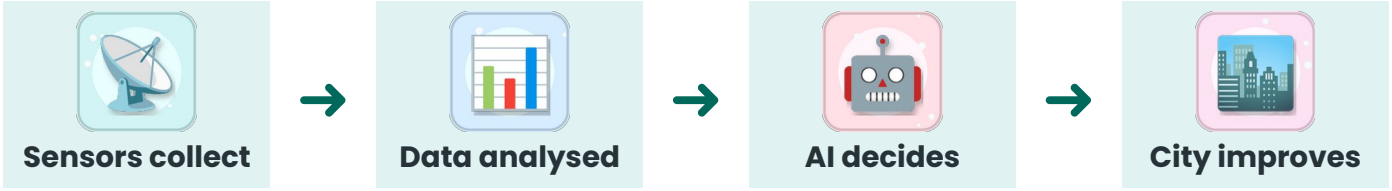
10:18

😊 Type a message...

📌 **Pinned by Bolt**

Smart cities use sensors, data and AI to manage traffic, water, power and waste. A truly smart city is accessible to all citizens.

🔄 How it works

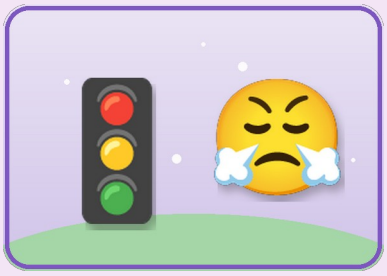


INDIA SPOTLIGHT

India's Smart Cities Mission set up command-and-control centres that use sensor data to manage traffic and city services.

AI CHRONICLES: The Signal That Listened

Kabir | Ms. Kavya | City Council

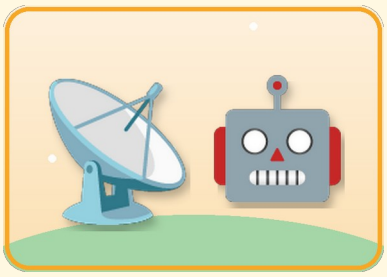


SCENE 1

Every morning, Kabir's school bus waited 5 minutes at a red signal on an empty road.

His team counted vehicles for a week. Traffic was heavy one way, empty the other.

SCENE 2



SCENE 3

They proposed a smart signal with sensors that gives green time based on real traffic.

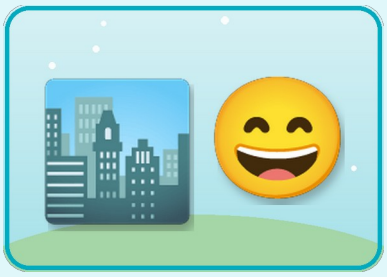
Ms. Kavya, a wheelchair user, added: "Include audio beeps and longer crossing time for us."

SCENE 4

SCENE 5

The city council tested the design. Waiting time fell and crossing became safer for all 🚶 .

"A city is smart only when it works for everyone," Kabir told the local newspaper 📰.



SCENE 6

🔮 What happens next? Tick ✓ one!

☐	☐	☐
Kabir's team designs smart bus stops	**The signal stays on fixed timers**	**The signal starts dancing**

🧐 Story check: answer the questions.

1. What problem did Kabir notice?

2. What did Ms. Kavya suggest?

3. What makes a city truly smart?

Rate this story: shade the stars.

☆ ☆ ☆ ☆ ☆

LOOK AROUND YOU: Smart city solutions

Metro crowd prediction

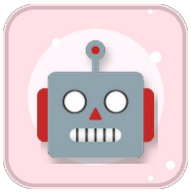
Parking finder

Street lights dim when empty

Bins that say "I'm full!"

Leak detection in pipes

Air quality boards



BOLT SAYS:
A city is smart when it helps every person: children, elders and people with disabilities!

FUN FACT!
India's Smart Cities Mission has worked on making over 100 cities smarter!

👀 **I spy AI! Draw one AI helper you saw today.**

LET'S DO IT! Design a Smart Street

You need:

Chart paper

Colours

Groups

Steps:

1		Draw one street of your town.
2		Add 4 sensors: traffic, light, bin, air.
3		Write what the AI does with each sensor's data.
4		Add one feature for elders or wheelchair users.

THINK & TALK

Which problem in your town would you fix first?

Draw what you did!

WORKSHEET: AI for Better Cities (Part A)

Name: _____ Date: _____

1. Match the city problem to the smart solution.

Traffic jams ●

Overflowing bins ●

Pipe leaks ●

Wasted street light ●

●

Lights dim when empty

●

Smart signals

●

Leak sensors

●

Bins that alert

2. Design a smart bus stop for your town!

My smart feature: _____

WORKSHEET: AI for Better Cities (Part B)

Name: _____ Date: _____

1. Circle the smart city features.

Smart signals

Bins that alert

Dinosaur park

Auto street lights

Air quality boards

Balloon shop

2. Fill in the blanks. Use the word box.

WORD BOX: sensor ✿ accessible ✿ smart city

A _____ collects city data.

Usable by everyone is _____.

A city using tech well is a _____.

PUZZLE FUN: AI for Better Cities

Name: _____ Date: _____

1. Odd one out! Circle it and justify your choice.

A				
B				
C				

2. Define it! Write each term in your own words with an example.

Sensor

Meaning: _____

Example: _____

Smart city

Meaning: _____

Example: _____

Accessible

Meaning: _____

Example: _____

🧪 DEMO LAB: Smart-city map design (with Google Maps traffic layer)

Name: _____ Date: _____

🎯 **Goal:** Study live traffic and design a smart, accessible neighbourhood.

🔗 **Open:** maps.google.com + chart paper | 👩‍🏫 **With a teacher or parent**

👣 Steps

1		Open the traffic layer for your town at 9 a.m. and at 2 p.m.
2		Mark the busiest roads in red on your sketch map.
3		Add 4 sensors: traffic, water, air, bins.
4		Add accessibility features: ramps, audio signals, shade.
5		Present your plan in 1 minute.

🔬 Record your observations

Busiest road:	
Sensors placed:	
Accessibility features added:	

💭 **Reflect:** How can a smart city make sure no citizen is left out?

? QUIZ TIME: AI for Better Cities

Circle the right answer!

Q1. Smart cities use sensors and...

a) Data & AI

b) Magic

c) Balloons

Q2. Usable by everyone means...

a) Blocked

b) Accessible

c) Costly

Q3. Smart bins tell when they are...

a) Sleepy

b) Singing

c) Full

Q4. Smart lights save...

a) Power

b) Pizza

c) Toys

Q5. India's mission for smarter cities is...

a) Moon Mission

b) Smart Cities Mission

c) Game Mission

MY SCORE: _____ / 5. Shade one star per correct answer.

☆ ☆ ☆ ☆ ☆

✅ TRUE OR FALSE: AI for Better Cities

Look at the picture. Tick ✓ TRUE or FALSE.

	👍 TRUE	👎 FALSE
Smart cities use sensors and AI.		
Good smart cities are for everyone.		
Smart lights waste more power.		
Smart bins can say when they're full.		
AI can predict metro crowds.		
Air quality doesn't matter in cities.		

HOME CONNECT

Walk your street with an adult. Note 3 problems a smart city could fix.

🔢 Self-assessment: tick one.

I know it well!

I know some.

I need practice.

EXERCISE: AI for Better Cities

Name: _____ Date: _____

A. Tick (✓) the correct answer.

1. Devices that collect city data are ____.

(a) sensors ☐ **(b)** posters ☐ **(c)** benches ☐

2. Smart signals adjust green time to ____.

(a) the weather ☐ **(b)** real traffic ☐ **(c)** music ☐

3. A city designed for everyone is ____.

(a) expensive ☐ **(b)** empty ☐ **(c)** accessible ☐

4. Smart bins tell trucks when they are ____.

(a) full ☐ **(b)** hungry ☐ **(c)** broken ☐

B. Fill in the blanks. Use the word box.

AI **wheelchair** **see** **Smart**

1. Smart cities use sensors, data and ____.

2. Ramps help ____ users.

3. Audio signals help people who cannot ____.

4. India's ____ Cities Mission upgrades cities.

C. Write T for True and F for False.

1. Sensors help cities collect data. [ ]

2. Smart cities only need to serve the rich. [ ]

3. AI can manage traffic signals. [ ]

4. Accessibility is not important in smart cities. [ ]

D. Match the following. Write the letter.

Column A		Column B
1. Sensor	____	**(a)** Reports when full
2. Smart signal	____	**(b)** Collects data
3. Smart bin	____	**(c)** Accessible design
4. Ramp	____	**(d)** Adjusts to traffic

E. Answer in one word.

1. What collects data in a smart city? _____

2. What design works for everyone? _____

3. Which mission upgrades Indian cities? _____

F. Answer these questions.

1. Name four city services AI can improve.

2. How can smart cities include everyone?

G. Case-based questions. Read and answer.

Kabir's team counted vehicles and proposed sensor-based signals. A wheelchair user suggested audio beeps and longer crossing times. Waiting fell and crossing became safer.

1. The team's data came from ____.

(a) counting vehicles ☐ **(b)** guessing ☐ **(c)** a movie ☐

2. The audio beeps make the signal more ____.

(a) noisy ☐ **(b)** accessible ☐ **(c)** expensive ☐

3. The result was ____.

(a) more waiting ☐ **(b)** no change ☐ **(c)** less waiting and safer crossing ☐

H. THINK DEEPER

Design one smart solution for a problem in your town. Which sensors would it need, and how would it help every citizen?

UNIT REVIEW: AI SOLVE

Name: _____ Date: _____

Circle the right answer!

Q1. Who decides treatment?

a) Doctor

b) Robot

c) App

Q2. Usable by everyone is...

a) Blocked

b) Accessible

c) Costly

Q3. Health data should be...

a) Public

b) Posted

c) Confidential

Q4. A sensor collects...

a) Data

b) Sweets

c) Toys

Q5. Doctor by video call is...

a) Gaming

b) Telemedicine

c) TV

Q6. Smart lights save...

a) Pizza

b) Toys

c) Power

WORD SEARCH: AI SOLVE

Find these words. They go across → or down ↓.

DIAGNOSIS

SENSOR

SMART CITY

ACCESSIBLE

DOCTOR

PRIVACY

N	U	S	E	N	S	O	R	P	E
M	D	B	E	R	G	I	O	K	O
S	P	R	I	V	A	C	Y	N	U
M	T	U	O	U	H	L	A	I	W
F	L	T	U	U	D	G	U	I	K
D	C	R	R	C	M	C	O	E	A
K	O	O	D	O	C	T	O	R	D
B	S	M	A	R	T	C	I	T	Y
A	C	C	E	S	S	I	B	L	E
W	D	I	A	G	N	O	S	I	S

I finished AI SOLVE! Teacher's sign: _____

MONTH 4

AI RISE

Skills • Careers • Future

Area: AI & Career

In this unit, you will explore:

Topic 7

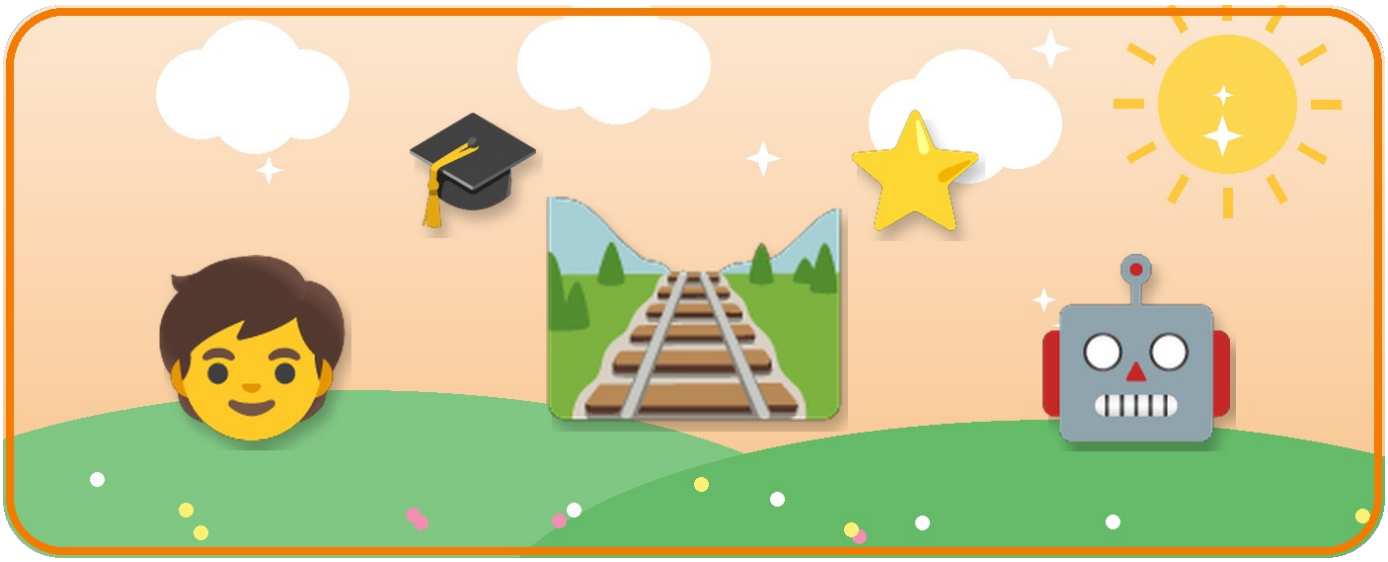
Find Your Future Path

Topic 8

AI Career Discovery

TOPIC 7 • MONTH 4 • AI & CAREER

Find Your Future Path



LET'S LEARN

- Your future path starts with your interests and strengths.
- Subjects in school connect to many careers.
- Explore, try, learn and keep adjusting your path!

Path builders

Interests

Strengths

Subjects

Goals

MAGIC WORDS

Pathway Route to a career

Stream Group of subjects

Goal What you aim for

💬 TECH TALK: Find Your Future Path

💬 GAIO Squad Riya, Kabir, Bolt • online

👧 **Riya**

Next year we choose streams 🤯. Science, commerce or humanities?

10:02

🤖 **Bolt**

Start with you: interests ❤️, strengths 💪 and values. Then explore pathways.

10:04 ✓✓

👦 **Kabir**

Does maths only lead to engineering?

10:06

🤖 **Bolt**

No! Maths leads to data science, economics, architecture 🏛️ and even music tech 🎵.

10:08 ✓✓

👧 **Riya**

And if I love biology and computers?

10:10

🤖 **Bolt**

Bioinformatics or health-tech AI 🧬! Subjects connect to many careers.

10:12 ✓✓

👦 **Kabir**

What if I change my mind later?

10:14

🤖 **Bolt**

That's normal. Set goals 🎯, try internships and clubs, and adjust your path 🔄.

10:16 ✓✓

👧 **Riya**

Explore, try, learn, adjust 🧭!

10:18

😊 Type a message...

📌 **Pinned by Bolt**

Your future path starts with your interests and strengths. School subjects connect to many careers. Set goals, explore and adjust as you grow.

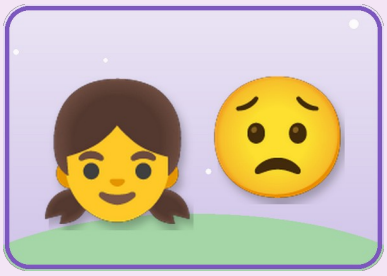
🔄 How it works

INDIA SPOTLIGHT

Many Indian state education departments and schools now offer career-guidance portals that map subjects to hundreds of careers.

AI CHRONICLES: The Stream Dilemma

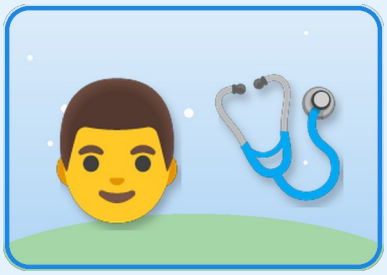
Riya | Dad | Counsellor Ms. Rose



SCENE 1

Riya loved biology and coding, but everyone said, "Pick one. You can't do both 😕."

Dad wanted her to be a doctor. Riya wasn't sure; she also loved building apps 📱.



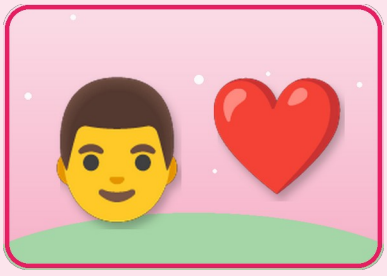
SCENE 2

SCENE 3

Ms. Rose showed her a pathway map: Biology + Computers Bioinformatics, Health-tech AI 🧬.

Riya joined a summer project analysing plant DNA data with Python. She loved it 😍!

SCENE 4



SCENE 5

Dad saw her excitement. "You can help doctors with technology. That's healing too."

Riya wrote her goal: 'Science stream, coding club, one research project a year.'

SCENE 6

🔮 What happens next? Tick ✓ one!

☐	☐	☐
Riya builds an AI health-app prototype	**Riya stops exploring**	**The pathway map gets lost**

🧐 Story check: answer the questions.

1. Why was Riya confused?

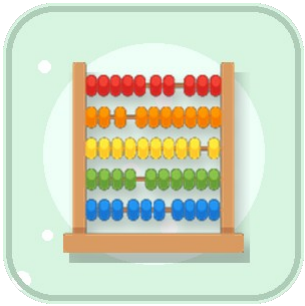
2. What pathway did Ms. Rose show?

3. What goal did Riya set?

Rate this story: shade the stars.

☆ ☆ ☆ ☆ ☆

LOOK AROUND YOU: Subjects → careers



Maths → data science, finance

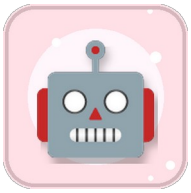
Science → medicine, research

Art → design, animation

Languages → media, law

Computers → software, AI

Social studies → policy, NGOs



BOLT SAYS:
There's no wrong path if you keep learning. Paths can change and grow!

FUN FACT!
After Class 10, students in India can choose streams like Science, Commerce, Arts or vocational courses!

I spy AI! Draw one AI helper you saw today.

LET'S DO IT! Interest Mapping

You need:

Interest map

Colours

Checklist

Steps:

1. List 5 things you enjoy doing.
2. Circle 3 things you're good at.
3. Link them to school subjects.
4. Find 2 careers that match. Draw your path!

THINK & TALK

Did your path surprise you?

Draw what you did!

WORKSHEET: Find Your Future Path (Part A)

Name: _____ Date: _____

1. Match the subject to a career.

Maths

Biology

Art

Computer Science

Animator

Doctor

AI engineer

Data scientist

2. Fill in the blanks. Use the word box.

WORD BOX: stream ✿ goal ✿ pathway

A route to a career is a _____.

A group of subjects is a _____.

What you aim for is a _____.

WORKSHEET: Find Your Future Path (Part B)

Name: _____ Date: _____

1. Interest or strength? Tick ✓ for you.

	Interest	Strength
Drawing		
Maths		
Speaking		
Computers		
Science		

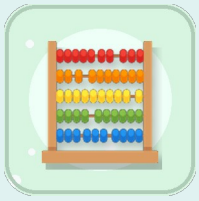
2. Draw your future path: today → Class 10 → career!

My goal: _____

PUZZLE FUN: Find Your Future Path

Name: _____ Date: _____

1. Odd one out! Circle it and justify your choice.

A				
B				
C				

2. Define it! Write each term in your own words with an example.

Pathway

Meaning: _____

Example: _____

Stream

Meaning: _____

Example: _____

Goal

Meaning: _____

Example: _____

🧪 DEMO LAB: Career pathway map (unplugged + online research with teacher)

Name: _____ Date: _____

🎯 **Goal:** Map your subjects to future streams and careers.

🔗 **Open:** chart paper + career portals | 👩‍🏫 **With a teacher or parent**

👣 Steps

1. List your 3 favourite school subjects.
2. For each, find 3 careers linked to it.
3. Find one career that combines two subjects (e.g., bio + computers).
4. Write one goal for this year (club, course or project).
5. Draw your pathway: subject → stream → course → career.

🔬 Record your observations

Subjects chosen:	
Combined career found:	
My goal this year:	

💭 **Reflect:** Why is it okay to change your path as you learn more?

? QUIZ TIME: Find Your Future Path

Circle the right answer!

Q1. A route to a career is a...

a) Pathway

b) Balloon

c) Cake

Q2. Biology can lead to...

a) Animation

b) Medicine

c) Farming only

Q3. A group of subjects is a...

a) River

b) Song

c) Stream

Q4. Can career paths change?

a) Yes

b) Never

c) Only at 50

Q5. Your path starts with your...

a) Shoes

b) Interests

c) TV

MY SCORE: ____ / 5. Shade one star per correct answer.

☆ ☆ ☆ ☆ ☆

✅ TRUE OR FALSE: Find Your Future Path

Look at the picture. Tick ✓ TRUE or FALSE.

		👍 TRUE	👎 FALSE
🛤️	A pathway is a route to a career.		
🔁	Career paths can change.		
🧮	Only maths students get good careers.		
🎓	Streams are groups of subjects.		
❤️	Interests help choose a path.		
🙈	You must decide your career at age 5.		

HOME CONNECT

Ask 3 family members how they chose their careers. Did their paths change?

🔢 Self-assessment: tick one.

I know it well!

I know some.

I need practice.

EXERCISE: Find Your Future Path

Name: _____ Date: _____

A. Tick (✓) the correct answer.

1. Choosing science, commerce or humanities is choosing a ____.

(a) stream ☐ **(b)** river ☐ **(c)** password ☐

2. A route from subjects to a career is a ____.

(a) password ☐ **(b)** pathway ☐ **(c)** playlist ☐

3. Biology + computers can lead to ____.

(a) cooking ☐ **(b)** tailoring ☐ **(c)** bioinformatics ☐

4. Changing your plan as you learn is ____.

(a) normal ☐ **(b)** wrong ☐ **(c)** impossible ☐

B. Fill in the blanks. Use the word box.

strengths goal science adjust

1. Know your interests and ____.

2. Set a clear ____ for the year.

3. Maths can lead to data ____.

4. Explore, try, learn and ____.

C. Write T for True and F for False.

1. One subject can lead to many careers. [ ]

2. You must never change your career plan. [ ]

3. Clubs and projects help you explore. [ ]

4. Interests don't matter in career choice. [ ]

D. Match the following. Write the letter.

Column A		Column B
1. Stream	____	**(a)** Target to achieve
2. Pathway	____	**(b)** Subject group after Class 10
3. Goal	____	**(c)** Work experience
4. Internship	____	**(d)** Route to a career

E. Answer in one word.

1. What is a subject group after Class 10? _____

2. What links subjects to careers? _____

3. Biology + computers career? _____

F. Answer these questions.

1. Map one of your favourite subjects to three careers.

2. Why is it okay to change your path?

G. Case-based questions. Read and answer.

Riya loves biology and coding. A counsellor shows her bioinformatics. After a summer project analysing DNA data with Python, she sets goals in science and coding.

1. Riya's interests are _____.

(a) biology and coding ☐ **(b)** only sports ☐ **(c)** none ☐

2. Bioinformatics combines _____.

(a) art and music ☐ **(b)** biology and computers ☐ **(c)** history and dance ☐

3. The summer project helped her _____.

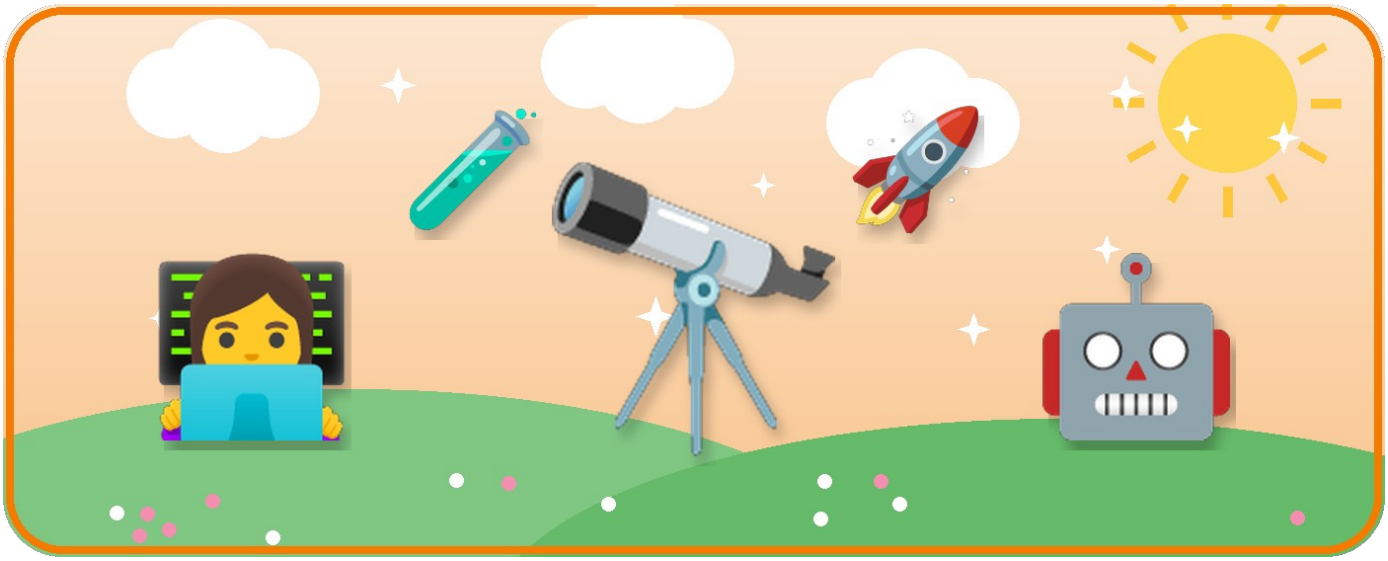
(a) give up ☐ **(b)** forget coding ☐ **(c)** test her interest ☐

H. THINK DEEPER

Create your own 5-step pathway from today to a dream career. Include one backup path.

TOPIC 8 • MONTH 4 • AI & CAREER

AI Career Discovery



LET'S LEARN

- AI careers are growing in every field.
- Some build AI, some use AI, some guide AI responsibly.
- Start now: learn maths, coding and clear communication!

Three AI career groups

Build AI

Use AI at work

Guide AI

Teach AI

MAGIC WORDS

ML Machine learning

Policy Rules & guidelines

Portfolio Collection of your work

💬 TECH TALK: AI Career Discovery

💬 **GAIO Squad** Riya, Kabir, Bolt • online

👦 **Kabir**
Bolt, what jobs exist in AI 💼?
10:02

🤖 **Bolt**
Three kinds: people who BUILD AI, people who USE AI and people who GUIDE AI.
10:04 ✓✓

👧 **Riya**
Builders?
10:06

🤖 **Bolt**
ML engineers 🛠️ train models; data scientists 📊 find insights; roboticists build robots 🦾.
10:08 ✓✓

👦 **Kabir**
Users?
10:10

🤖 **Bolt**
Doctors, farmers, designers and teachers using AI tools every day 👩‍🌾.
10:12 ✓✓

Riya
And guides?
10:14

Bolt
AI policy experts and ethicists ⚖️ who make rules so AI is safe and fair.
10:16 ✓✓

Kabir
How do we start now?
10:18

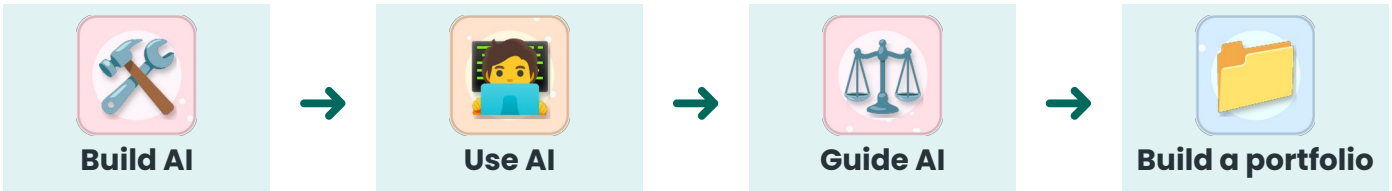
Bolt
Learn maths, coding and communication, and build a portfolio of projects 📁!
10:20 ✓✓

😊 Type a message...

📌 **Pinned by Bolt**

AI careers include building AI (ML engineer), using AI (in every field) and guiding AI (policy, ethics). Start with maths, coding, communication and a project portfolio.

🔄 How it works

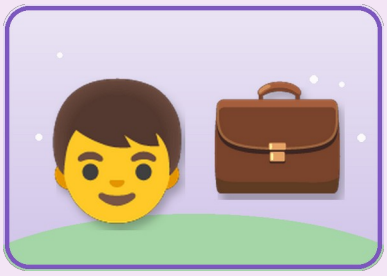


INDIA SPOTLIGHT

India's IndiaAI Mission aims to grow AI skills, computing power and start-ups so more young people can build AI.

AI CHRONICLES: The Portfolio Party

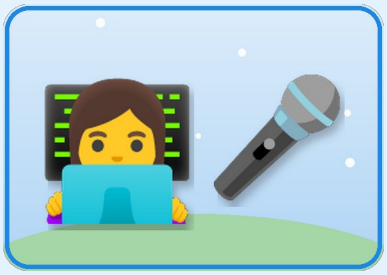
Kabir | Engineer Priya | Bolt



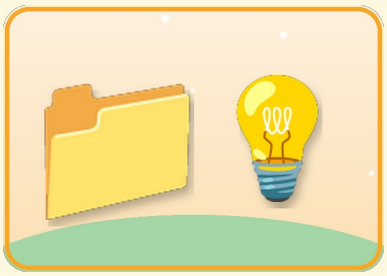
SCENE 1

Kabir thought AI jobs were only for geniuses with big degrees 🎓.

At a school talk, engineer Priya said, "I was hired because of my projects, not just marks."



SCENE 2

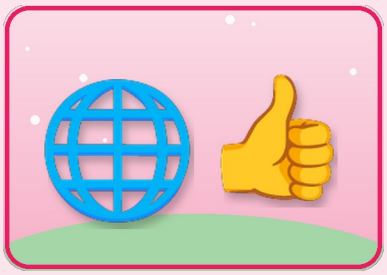


SCENE 3

She explained a portfolio: a collection of things you've built, like apps, datasets or robots.

Kabir started small: a spam checker, a mango-sorter model and a smart signal sketch 📊.

SCENE 4



SCENE 5

He shared them on the school showcase page, with a short write-up for each project.

Months later, a local start-up invited him for a summer internship 🎉!

SCENE 6

🔮 What happens next? Tick ✓ one!

☐	☐	☐
Kabir mentors juniors on portfolios	**Kabir deletes his projects**	**The portfolio throws a real party**

🧐 Story check: answer the questions.

1. What did Kabir believe at first?

2. What is a portfolio?

3. What happened because of his portfolio?

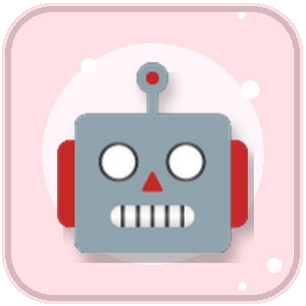
Rate this story: shade the stars.

☆ ☆ ☆ ☆ ☆

LOOK AROUND YOU: AI careers to discover

ML engineer

Data analyst

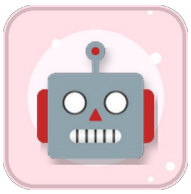


Robotics engineer

AI product designer

AI policy expert

AI security expert



BOLT SAYS:

Builders, users and guides: the AI world needs all three!

FUN FACT!

Online courses let students anywhere learn AI basics for free!

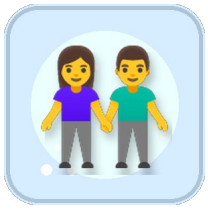
I spy AI! Draw one AI helper you saw today.

LET'S DO IT! AI Career Interview Role-Play

You need:

Career cards

Toy mic



Partners

Steps:

1. Pick an AI career card.
2. Research: what they do, skills, a day at work.
3. One partner interviews the other as that expert.
4. Add your notes to your career portfolio!

THINK & TALK

Which AI career group suits you: build, use or guide?

Draw what you did!

WORKSHEET: AI Career Discovery (Part A)

Name: _____ Date: _____

1. Build AI or Use AI? Tick ✓.

	Builds AI	Uses AI
ML engineer		
Doctor with AI scanner		
Robotics engineer		
Farmer with crop app		
AI data engineer		

2. Make a portfolio page: your dream AI career!

Career: _____

WORKSHEET: AI Career Discovery (Part B)

Name: _____ Date: _____

1. Match the AI career to the group.

ML engineer ● ● Guides AI

Farmer with AI app ● ● Builds AI

Policy expert ● ● Teaches AI

AI teacher ● ● Uses AI

2. Fill in the blanks. Use the word box.

WORD BOX: portfolio ✿ policy ✿ ML

Machine learning is called _____.

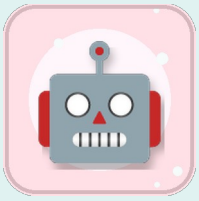
A collection of your work is a _____.

Rules and guidelines are _____.

PUZZLE FUN: AI Career Discovery

Name: _____ Date: _____

1. Odd one out! Circle it and justify your choice.

A				
B				
C				

2. Define it! Write each term in your own words with an example.

ML

Meaning: _____

Example: _____

Policy

Meaning: _____

Example: _____

Portfolio

Meaning: _____

Example: _____

🧪 DEMO LAB: Online job portal / career site (with teacher)

Name: _____ Date: _____

🎯 **Goal:** Discover real AI jobs and the skills they need.

🔗 **Open:** any job or career portal | 👩‍🏫 **With a teacher or parent**

👣 Steps

1. Search 'AI engineer', 'data analyst' and 'AI policy'.
2. Pick 2 job posts. List the skills they ask for.
3. Tick skills you already have (maths, coding, teamwork...).
4. Plan 2 projects for your portfolio.
5. Share your top 3 skills to build in a 1-minute talk.

🔬 Record your observations

Most common skill required:	
Skill I already have:	
Portfolio project idea:	

💭 **Reflect:** Why do AI careers need communication and ethics, not just coding?

? QUIZ TIME: AI Career Discovery

Circle the right answer!

Q1. ML stands for...

a) Machine learning

b) More lemons

c) Music list

Q2. A collection of your work is a...

a) Bag

b) Portfolio

c) Lunch box

Q3. Who guides AI rules?

a) Gamer

b) Chef

c) Policy expert

Q4. A doctor using an AI scanner...

a) Uses AI

b) Builds AI

c) Ignores AI

Q5. To start an AI career, learn...

a) Nothing

b) Maths & coding

c) Only TV

MY SCORE: ____ / 5. Shade one star per correct answer.

☆ ☆ ☆ ☆ ☆

✅ TRUE OR FALSE: AI Career Discovery

Look at the picture. Tick ✓ TRUE or FALSE.

		👍 TRUE	👎 FALSE
🛠️	Some careers build AI.		
👩‍⚕️	Doctors can use AI at work.		
💻	AI careers are only for coders.		
⚖️	Policy experts help guide AI.		
🌍	Online courses can teach AI basics.		
😴	You don't need to learn anything for AI careers.		

HOME CONNECT

Interview a relative: does AI help in their work? Add it to your portfolio.

🔢 Self-assessment: tick one.

I know it well!

I know some.

I need practice.

EXERCISE: AI Career Discovery

Name: _____ Date: _____

A. Tick (✓) the correct answer.

1. A person who trains AI models is an ____.

(a) ML engineer ☐ **(b)** MC host ☐ **(c)** MP ☐

2. Experts who make rules for safe AI work in AI ____.

(a) painting ☐ **(b)** policy ☐ **(c)** plumbing ☐

3. A collection of your projects is a ____.

(a) passport ☐ **(b)** playlist ☐ **(c)** portfolio ☐

4. Doctors using AI tools are AI ____.

(a) users ☐ **(b)** builders ☐ **(c)** breakers ☐

B. Fill in the blanks. Use the word box.

learning **insights** **fair** **IndiaAI**

1. ML stands for machine ____.

2. Data scientists find ____ in data.

3. AI ethicists keep AI ____.

4. India's ____ Mission grows AI skills.

C. Write T for True and F for False.

1. AI careers exist in every field. [ ]

2. Only coders can work in AI. [ ]

3. A portfolio shows your projects. [ ]

4. Communication skills don't matter in AI jobs. [ ]

D. Match the following. Write the letter.

Column A		Column B
1. ML engineer	____	(a) Makes AI rules
2. Data scientist	____	(b) Trains AI models
3. AI policy expert	____	(c) Builds robots
4. Roboticist	____	(d) Finds insights

E. Answer in one word.

1. What does ML stand for? _____
2. What shows your projects? _____
3. Who makes AI rules? _____

F. Answer these questions.

1. Explain the three kinds of AI careers: build, use, guide.

2. List three skills to start building now.

G. Case-based questions. Read and answer.

Kabir thought AI jobs needed big degrees. After an engineer's talk, he built small projects, shared them on a showcase page and got an internship offer.

1. Kabir's projects formed his ____.

(a) portfolio ☐ **(b)** passport ☐ **(c)** password ☐

2. The engineer was hired because of her ____.

(a) phone ☐ **(b)** projects ☐ **(c)** car ☐

3. Kabir's result was ____.

(a) a fine ☐ **(b)** nothing ☐ **(c)** an internship offer ☐

H. THINK DEEPER

Choose an AI career. Describe one day in that job and the skills needed for each task.

UNIT REVIEW: AI RISE

Name: _____ Date: _____

Circle the right answer!

Q1. A route to a career is a...

a) Pathway

b) Balloon

c) Cake

Q2. A collection of your work is a...

a) Bag

b) Portfolio

c) Lunch box

Q3. A group of subjects is a...

a) River

b) Song

c) Stream

Q4. ML stands for...

a) Machine learning

b) More lemons

c) Music list

Q5. Can career paths change?

a) Never

b) Yes

c) Only at 50

Q6. Who guides AI rules?

a) Gamer

b) Chef

c) Policy expert

WORD SEARCH: AI RISE

🔍 **Find these words. They go across → or down ↓.**

PATHWAY

STREAM

GOAL

ML

POLICY

PORTFOLIO

G	O	P	O	L	I	C	Y	O	W
K	P	O	R	T	F	O	L	I	O
O	G	O	A	L	P	F	H	K	I
B	C	B	P	I	S	T	R	L	E
G	C	O	G	P	I	F	M	M	O
U	L	S	T	R	E	A	M	L	T
G	L	D	B	H	I	U	W	H	D
L	F	K	P	A	B	M	C	K	L
A	L	P	A	T	H	W	A	Y	K
L	E	O	W	C	K	W	G	P	K

I finished AI RISE! Teacher's sign: _____

MONTH 5

AI CREATE

Try • Build • Innovate

Area: AI Tools

In this unit, you will explore:

Topic 9

Build with GenAI

Topic 10

AI Media Studio

TOPIC 9 • MONTH 5 • AI TOOLS

Build with GenAI

LET'S LEARN

- GenAI creates new text, images, code and music.
- It follows your prompt. Clear prompts = better results.
- Build ideas with it, then check and improve!

Prompt recipe

Role

Task

Details

Format

MAGIC WORDS

GenAI AI that creates

Prompt Instruction to AI

Iterate Improve again & again

💬 TECH TALK: Build with GenAI

💬 GAIO Squad Riya, Kabir, Bolt • online

👧 Riya
Bolt, can GenAI help build a real project 🛠️?
10:02

🤖 Bolt
Yes! GenAI creates text, images, code 💻 and music from prompts.
10:04 ✓✓

👦 Kabir
What makes a strong prompt?
10:06

🤖 Bolt
Role + task + context + format 📝. 'You are a science teacher. Make 5 quiz questions on magnets for Class 8, with answers.'
10:08 ✓✓

👧 Riya
And if the result isn't good?
10:10

🤖 Bolt
Iterate 🔁: refine the prompt, add examples, ask for changes.
10:12 ✓✓

👦 **Kabir**

Can it write code for my app?

10:14

🤖 **Bolt**

It can draft code, but you must test and understand it 🧪. AI code can have bugs.

10:16 ✓✓

👧 **Riya**

Prompt, create, check, improve 🚀!

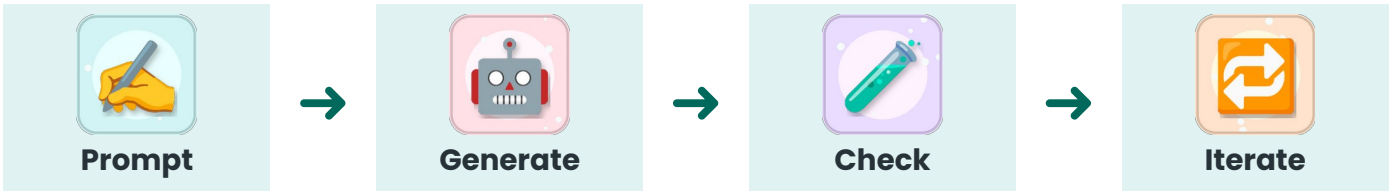
10:18

😊 Type a message...

📌 **Pinned by Bolt**

GenAI creates text, images, code and music from prompts. Use role, task, context and format; then check and iterate.

🔄 How it works



INDIA SPOTLIGHT

Indian start-ups are building GenAI models that understand Indian languages, so more people can use AI in their mother tongue.

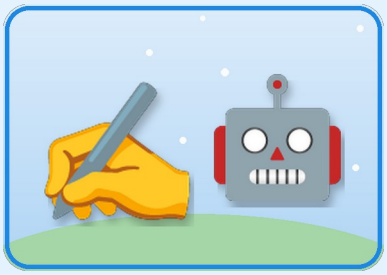
AI CHRONICLES: The App That Built Itself (Almost)

Riya Kabir GenAI

SCENE 1

The science teacher needed a quick revision quiz app. Riya and Kabir volunteered 🙋.

Their first prompt, 'make quiz', gave 5 random questions on history. Wrong subject 😅!



SCENE 2

SCENE 3

They rewrote it: role, task, context, format. Now they got 10 perfect magnet questions.

Checking with the textbook, they found one wrong answer. GenAI had made a mistake!

SCENE 4

SCENE 5

GenAI drafted Python code. It crashed on wrong inputs, so they added checks 🔧.

The class loved the app. "GenAI helped," said Kabir, "but we checked and improved."

SCENE 6

🔮 What happens next? Tick ✓ one!

☐	☐	☐
They add a leaderboard feature	**They stop checking AI output**	**The app writes its own exam**

🧐 Story check: answer the questions.

1. What was wrong with the first prompt?

2. What mistake did they find?

3. Why is checking important?

Rate this story: shade the stars.

☆ ☆ ☆ ☆ ☆

LOOK AROUND YOU: Things students build with GenAI

Quiz generators

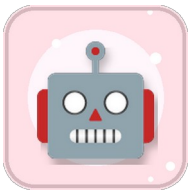
Story outlines

Concept art

Starter code

Study planners

Debate practice



BOLT SAYS:

Role + Task + Details + Format = a powerful prompt!

FUN FACT!

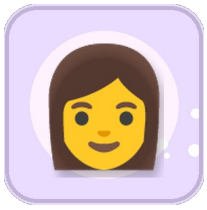
Changing just a few words in a prompt can completely change what GenAI creates!

I spy AI! Draw one AI helper you saw today.

LET'S DO IT! Prompt Upgrade Lab

You need:

Prompt sheet



Teacher's AI tool

Notebook

Steps:

1		Weak: "Make a quiz."
2		Add role & task: "You are a science teacher. Make a quiz."
3		Add details & format: "5 MCQs on plants, for Class 8."
4		Compare results. Which is best? Why?

THINK & TALK

Which part of the prompt recipe made the biggest difference?

Draw what you did!

WORKSHEET: Build with GenAI (Part A)

Name: _____ Date: _____

1. Match the prompt part to the example.

Role ●		●	as a table
Task ●		●	You are a coach
Details ●		●	Make a plan
Format ●		●	for Class 8, 2 weeks

2. Weak or strong prompt? Tick ✓.

	Strong	Weak
As a teacher, make 5 MCQs on plants		
Do something		
Explain gravity in 3 bullet points		
Write stuff		

WORKSHEET: Build with GenAI (Part B)

Name: _____ Date: _____

1. **Building with GenAI. Write 1, 2, 3, 4.**

Check the output ☐

Plan your idea ☐

Improve & iterate ☐

Write the prompt ☐

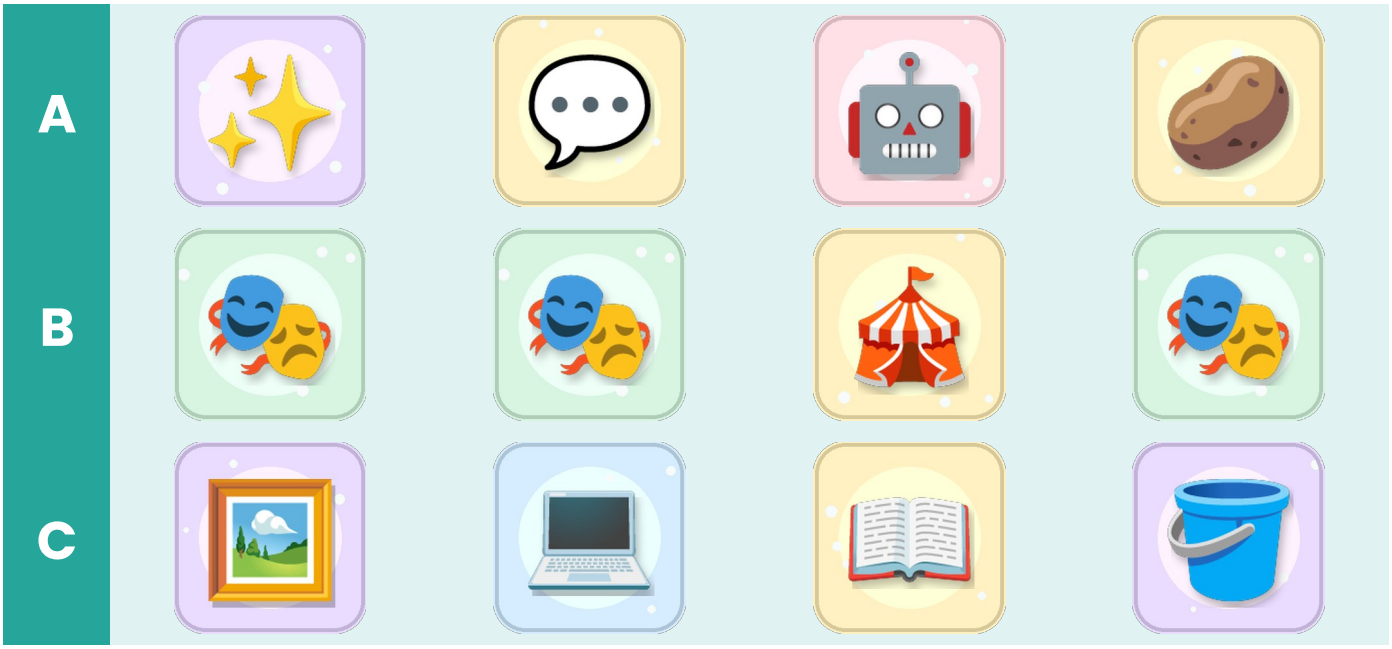
2. **Write a 4-part prompt (Role, Task, Details, Format) for a study plan!**

My prompt: _____

PUZZLE FUN: Build with GenAI

Name: _____ Date: _____

1. Odd one out! Circle it and justify your choice.



2. Define it! Write each term in your own words with an example.

GenAI

Meaning: _____

Example: _____

Prompt

Meaning: _____

Example: _____

Iterate

Meaning: _____

Example: _____

🧪 DEMO LAB: A school-approved GenAI chatbot (with teacher)

Name: _____ Date: _____

🎯 **Goal:** Build a mini quiz app using GenAI prompts, then check it.

🔗 **Open:** any approved chatbot | 👩‍🏫 **With a teacher or parent**

👣 Steps

1		Prompt: 'You are a teacher. Make 5 MCQs on magnets for Class 8 with answers.'
2		Check every answer with your textbook.
3		Iterate: 'Make question 3 harder and add a hint.'
4		Ask for simple Python code to run the quiz.
5		Run the code. Fix any bug you find.

🔬 Record your observations

Errors found in AI answers:	
How iterating changed the quiz:	
Bug found in AI code:	

💭 **Reflect:** Why must we check GenAI output before using it?

? QUIZ TIME: Build with GenAI

Circle the right answer!

Q1. AI that creates new content is...

a) GenAI

b) CleanAI

c) SleepAI

Q2. Improving again and again is to...

a) Stop

b) Iterate

c) Delete

Q3. "You are a coach" is the...

a) Format

b) Task

c) Role

Q4. "as a table" is the...

a) Format

b) Role

c) Toy

Q5. Clear prompts give...

a) Worse results

b) Better results

c) No results

MY SCORE: ____ / 5. Shade one star per correct answer.

☆ ☆ ☆ ☆ ☆

✅ TRUE OR FALSE: Build with GenAI

Look at the picture. Tick ✓ TRUE or FALSE.

		👍 TRUE	👎 FALSE
✨	GenAI can create text, images and code.		
💬	Clear prompts give better results.		
🔍	GenAI output never needs checking.		
🎭	Role is part of the prompt recipe.		
🔁	Iterate means improve again and again.		
🤷	"Do something" is a strong prompt.		

HOME CONNECT

With a parent, write a GenAI prompt for a family trip plan. Check the plan together!

🔢 Self-assessment: tick one.

I know it well!

I know some.

I need practice.

EXERCISE: Build with GenAI

Name: _____ Date: _____

A. Tick (✓) the correct answer.

1. AI that creates new content is ____.

(a) GenAI ☐ **(b)** GPS ☐ **(c)** GIF ☐

2. A strong prompt includes role, task, context and ____.

(a) font ☐ **(b)** format ☐ **(c)** fee ☐

3. Improving output step by step is to ____.

(a) ignore ☐ **(b)** delete ☐ **(c)** iterate ☐

4. GenAI-written code must be ____.

(a) tested ☐ **(b)** trusted blindly ☐ **(c)** printed ☐

B. Fill in the blanks. Use the word box.

prompt **textbook** **bugs** **better**

1. GenAI follows your ____.

2. Check answers against your ____.

3. AI code can contain ____.

4. Clear prompts give ____ results.

C. Write T for True and F for False.

1. GenAI can create code. [ ]

2. GenAI never makes mistakes. [ ]

3. Adding context improves prompts. [ ]

4. You should never refine a prompt. [ ]

D. Match the following. Write the letter.

Column A		Column B
1. Role	____	**(a)** Background details
2. Task	____	**(b)** Who AI should act as
3. Context	____	**(c)** How to present it
4. Format	____	**(d)** What to do

E. Answer in one word.

1. What is AI that creates content called? _____

2. What do we give GenAI? _____

3. What is refining step by step? _____

F. Answer these questions.

1. Write a role-task-context-format prompt for a science poster.

2. Why must we test GenAI code?

G. Case-based questions. Read and answer.

'Make quiz' gave random history questions. A role-task-context-format prompt gave perfect magnet questions, but one answer was wrong and the code crashed on bad input.

1. The first prompt lacked _____.

(a) context ☐ **(b)** letters ☐ **(c)** length ☐

2. The wrong answer was caught by _____.

(a) guessing ☐ **(b)** checking the textbook ☐ **(c)** luck ☐

3. The crash was fixed by _____.

(a) deleting the app ☐ **(b)** ignoring it ☐ **(c)** adding input checks ☐

H. THINK DEEPER

Should students be allowed to use GenAI for projects? Suggest three fair rules.

TOPIC 10 • MONTH 5 • AI TOOLS

AI Media Studio

LET'S LEARN

- AI media tools edit videos, clean audio and create images.
- Creators use them for school projects, ads and films.
- Ethical creators label AI media and respect copyright!

Studio tools

Video editing

Audio cleanup

Image creation

Auto subtitles

MAGIC WORDS

Copyright
Creator's ownership

Label Mark AI content

Storyboard Plan of scenes

💬 TECH TALK: AI Media Studio

💬 GAIO Squad Riya, Kabir, Bolt • online

👦 Kabir
I'm making a school documentary 🎬. Can AI help?
10:02

🤖 Bolt
Yes! AI tools remove background noise 🎧, add subtitles and suggest edits.
10:04 ✓✓

👧 Riya
Where do we start?
10:06

🤖 Bolt
With a storyboard 🗂️: a scene-by-scene plan with pictures and notes.
10:08 ✓✓

👦 Kabir
Can I use any song from the internet?
10:10

🤖 Bolt
No. That breaks copyright ©. Use royalty-free music or create your own.
10:12 ✓✓

👧 **Riya**

And AI-made images?

10:14

🤖 **Bolt**

Label them: 'AI-generated' 🏷️. Viewers deserve to know what is real.

10:16 ✓✓

👦 **Kabir**

Plan, create, label, respect ✅!

10:18

😊 Type a message...

📌 **Pinned by Bolt**

AI media tools help edit video, clean audio and create images. Plan with a storyboard, respect copyright and label AI-made media.

🔄 How it works

INDIA SPOTLIGHT

India produces more films than almost any other country, and studios increasingly use AI for dubbing, subtitles and visual effects.

AI CHRONICLES: Lights, Camera, Label!

Kabir Riya Judge

SCENE 1

The district short-film contest theme was 'Our River' 🏞️. Kabir's team had one week.

Riya drew a 6-scene storyboard: clean river, pollution, cleanup drive, hope.

SCENE 2

SCENE 3

AI removed traffic noise from interviews and added subtitles in Tamil and English 🔤.

Kabir wanted a famous film song. Riya stopped him: "Copyright! Let's compose our own 🎹."

SCENE 4

SCENE 5

For an AI-made scene of the future river, they added a label: 'AI-generated image'.

The judge praised their honesty and creativity.
They won first prize 🏆 !

SCENE 6

🔮 What happens next? Tick ✓ one!

☐	☐	☐
The team makes a film on plastic waste	**They copy a film song next time**	**The river makes its own film**

🧐 Story check: answer the questions.

1. How did AI help the team?

2. Why didn't they use the film song?

3. What did they label and why?

Rate this story: shade the stars.

☆ ☆ ☆ ☆ ☆

LOOK AROUND YOU: AI in media today

Film special effects

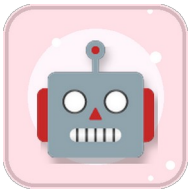
News graphics

Background music

Dubbing in many languages

Photo enhancement

Game characters



BOLT SAYS:

Label it! "AI-generated image" builds trust with your audience.

FUN FACT!

AI dubbing can help one film reach audiences in Tamil, Hindi, Telugu and more!

I spy AI! Draw one AI helper you saw today.

LET'S DO IT! Storyboard Studio

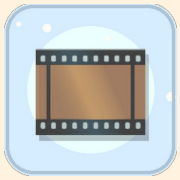
You need:

6-box sheet

Colours

Voice recorder

Steps:

1. Pick a topic: "Save Electricity".
2.  Draw a 6-scene storyboard.
3. Record a short voice-over (with teacher).
4. Label any AI images or music you used.

THINK & TALK

Why should AI-made media be labelled?

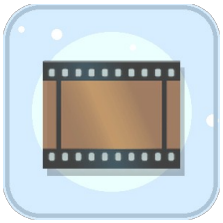
Draw what you did!

WORKSHEET: AI Media Studio (Part A)

Name: _____ Date: _____

1. Making a short film. Write 1, 2, 3, 4.

Label & share ☐



Storyboard ☐

Edit ☐

Shoot ☐

2. Fill in the blanks. Use the word box.

WORD BOX: copyright ✿ storyboard ✿ label

A plan of scenes is a _____.

A creator's ownership is _____.

I _____ AI-made images.

3. Bonus: Draw your favourite picture from this topic!

I drew _____

WORKSHEET: AI Media Studio (Part B)

Name: _____ Date: _____

1. Match the studio tool to its job.

Video editor ●

Audio cleaner ●

Auto subtitles ●

Image maker ●

●

Adds text

●

Creates pictures

●

Cuts clips

●

Removes noise

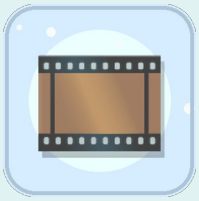
2. Ethical or not? Tick ✓.

	Ethical	Not ethical
Labelling AI images		
Using songs without permission		
Adding subtitles for all		
Faking a friend's voice		

PUZZLE FUN: AI Media Studio

Name: _____ Date: _____

1. Odd one out! Circle it and justify your choice.

A				
B				
C				

2. Define it! Write each term in your own words with an example.

Copyright

Meaning: _____

Example: _____

Label

Meaning: _____

Example: _____

Storyboard

Meaning: _____

Example: _____

🧪 DEMO LAB: Free video editor with AI features (with teacher)

Name: _____ Date: _____

🎯 **Goal:** Plan and make a 1-minute AI-assisted video, ethically.

🔗 **Open:** any school-approved editor | 👩‍🏫 **With a teacher or parent**

👣 Steps

1		Draw a 6-box storyboard on 'Save Water'.
2		Record 4 short clips on a phone.
3		Use AI noise removal and auto-subtitles.
4		Add royalty-free music only.
5		Add an end card: 'AI tools used: ...' with credits.

🔬 Record your observations

AI features used:	
Music source:	
What I labelled:	

💭 **Reflect:** Why should creators label AI-made media?

QUIZ TIME: AI Media Studio

Circle the right answer!

Q1. A plan of scenes is a...

a) Storyboard

b) Cake

c) Balloon

Q2. AI media should be...

a) Hidden

b) Labelled

c) Deleted

Q3. Creator's ownership is...

a) Music

b) Pizza

c) Copyright

Q4. AI dubbing helps films reach...

a) More languages

b) The Moon

c) Fish

Q5. Auto subtitles add...

a) Music

b) Text of speech

c) Colour

MY SCORE: ____ / 5. Shade one star per correct answer.

✅ TRUE OR FALSE: AI Media Studio

Look at the picture. Tick ✓ TRUE or FALSE.

	👍 TRUE	👎 FALSE
A storyboard plans scenes.		
AI media should be labelled.		
Copyright means you can use anyone's work freely.		
AI dubbing helps films reach more languages.		
Faking someone's voice is fine.		
AI can clean up audio.		

HOME CONNECT

Make a 1-minute family video with a storyboard. Label any AI tools you used.

🔢 Self-assessment: tick one.

I know it well!

I know some.

I need practice.

EXERCISE: AI Media Studio

Name: _____ Date: _____

A. Tick (✓) the correct answer.

1. A scene-by-scene visual plan is a ____.

(a) storyboard ☐ **(b)** scoreboard ☐ **(c)** keyboard ☐

2. Using a song without permission breaks ____.

(a) grammar ☐ **(b)** copyright ☐ **(c)** gravity ☐

3. AI-made images should be ____.

(a) hidden ☐ **(b)** sold ☐ **(c)** labelled ☐

4. AI can remove background ____ from audio.

(a) noise ☐ **(b)** colour ☐ **(c)** light ☐

B. Fill in the blanks. Use the word box.

royalty **subtitles** **real** **creators**

1. Use ____-free music.

2. AI can add ____ in many languages.

3. Viewers deserve to know what is ____.

4. Credit tools and ____.

C. Write T for True and F for False.

1. AI tools can help edit videos. [ ]

2. Any internet song is free to use. [ ]

3. Storyboards help plan videos. [ ]

4. Labelling AI media is dishonest. [ ]

D. Match the following. Write the letter.

Column A		Column B
1. Storyboard	____	**(a)** Shows AI was used
2. Copyright	____	**(b)** Scene plan
3. Label	____	**(c)** On-screen text
4. Subtitles	____	**(d)** Creator's ownership

E. Answer in one word.

1. What is a scene-by-scene plan? _____

2. What protects a creator's work? _____

3. What tag shows AI media? _____

F. Answer these questions.

1. List four ways AI helps media creators.

2. Why should AI-made media be labelled?

G. Case-based questions. Read and answer.

A team used AI to clean audio and add subtitles, composed their own music instead of a film song, and labelled an AI-made scene. They won first prize.

1. Composing their own music respected ____.

(a) copyright ☐ **(b)** traffic rules ☐ **(c)** gravity ☐

2. The AI image was ____.

(a) hidden ☐ **(b)** labelled ☐ **(c)** deleted ☐

3. AI helped with ____.

(a) acting ☐ **(b)** the trophy ☐ **(c)** audio and subtitles ☐

H. THINK DEEPER

Design a storyboard for a 1-minute video on online safety. Mark where AI tools could help.

UNIT REVIEW: AI CREATE

Name: _____ Date: _____

Circle the right answer!

Q1. AI that creates is...

a) GenAI

b) CleanAI

c) SleepAI

Q2. A plan of scenes is a...

a) Cake

b) Storyboard

c) Balloon

Q3. "You are a coach" is the...

a) Format

b) Task

c) Role

Q4. AI media should be...

a) Labelled

b) Hidden

c) Deleted

Q5. Improving again and again is to...

a) Stop

b) Iterate

c) Delete

Q6. Creator's ownership is...

a) Music

b) Pizza

c) Copyright

WORD SEARCH: AI CREATE

Find these words. They go across → or down ↓.

GENAI	PROMPT	ITERATE	COPYRIGHT	LABEL	STORYBOARD

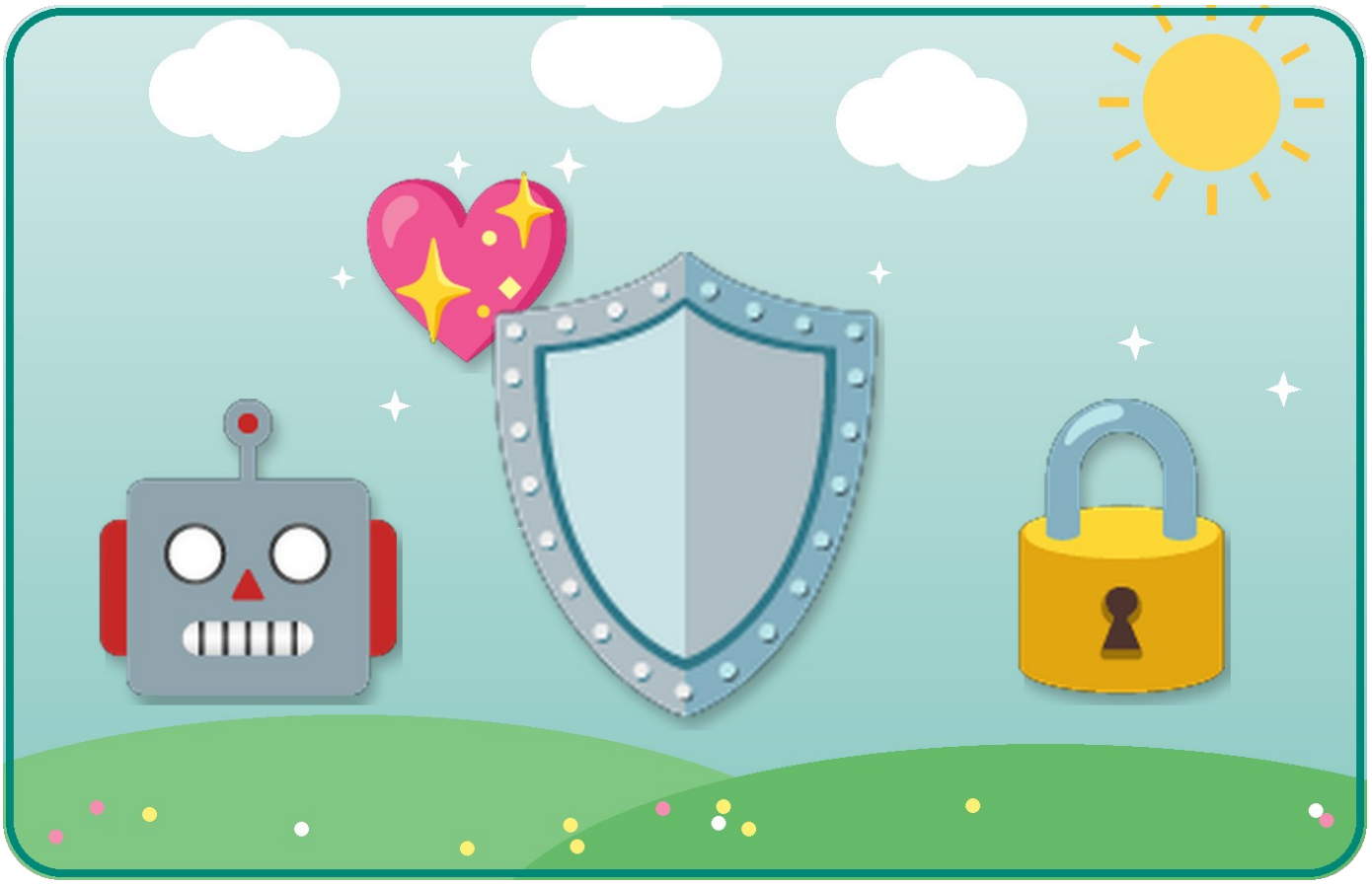
F	C	P	E	I	C	L	K	S	W
U	O	R	A	W	L	C	K	T	M
K	P	O	R	L	F	R	R	O	F
B	Y	M	G	E	N	A	I	R	I
A	R	P	M	N	A	T	O	Y	M
N	I	T	L	A	B	E	L	B	U
A	G	P	B	F	W	G	D	O	H
P	H	M	S	M	S	I	P	A	D
U	T	M	K	B	O	C	G	R	L
S	I	T	E	R	A	T	E	D	W

I finished AI CREATE! Teacher's sign: _____

MONTH 6

AI CARE

Verify • Protect • Respect



Area: Responsible AI

In this unit, you will explore:

Topic 11

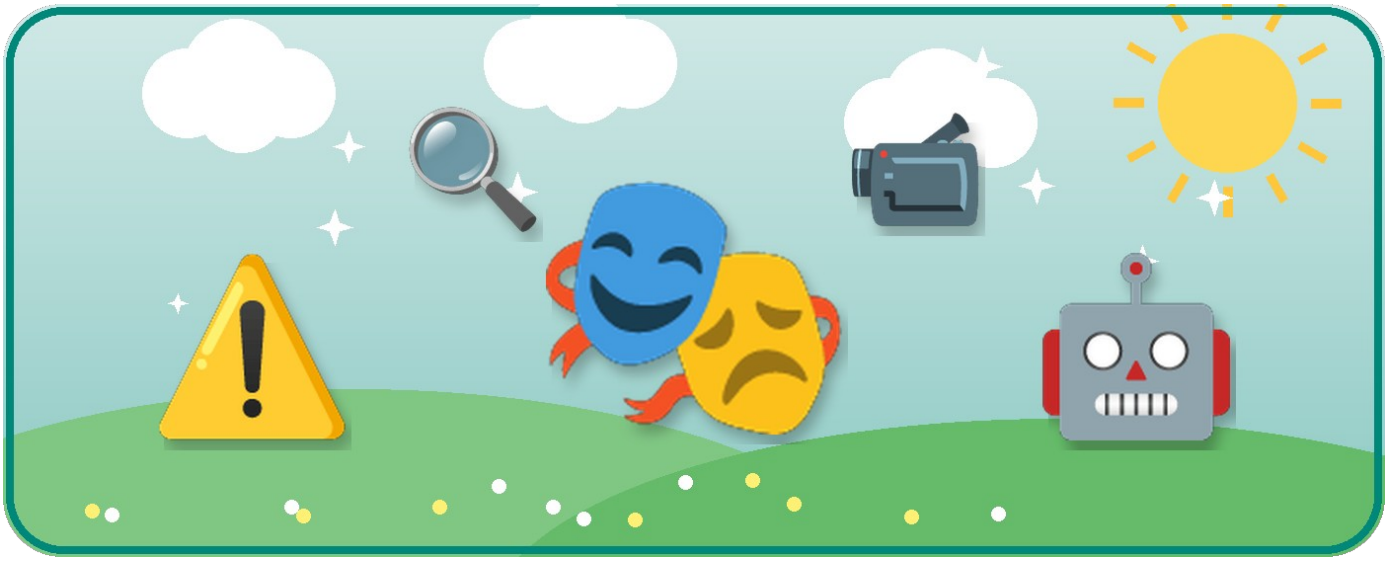
Deepfake Alert!

Topic 12

Your Data, Your Right

TOPIC 11 • MONTH 6 • RESPONSIBLE AI

Deepfake Alert!



LET'S LEARN

- Deepfakes are AI-made fakes of real people.
- They can spread lies or scam people.
- Spot, stop, check and report!

Deepfake warning signs

Lips don't match

Odd blinking

Strange lighting

Robotic voice

MAGIC WORDS

Deepfake AI-made fake media

Red flag Warning sign

Report Tell the platform/adult

💬 TECH TALK: Deepfake Alert!

💬 GAIO Squad Riya, Kabir, Bolt • online

👧 **Riya**
My aunt got a voice call that sounded exactly like her son asking for money 📞!
10:02

🤖 **Bolt**
That might be a voice-clone deepfake 🎭. AI can copy voices from short clips.
10:04 ✓✓

👦 **Kabir**
Scary! How do we spot them?
10:06

🤖 **Bolt**
Red flags 🚩: urgency, requests for money or OTP, odd pauses, unnatural faces or lips.
10:08 ✓✓

👧 **Riya**
What should she do?
10:10

🤖 **Bolt**
Stop 🛑, hang up and call the person back on their known number.
10:12 ✓✓

👦 **Kabir**

And deepfake videos online?

10:14

🤖 **Bolt**

Don't share 🚫. Report them on the platform or at cybercrime.gov.in.

10:16 ✓✓

👧 **Riya**

Spot, stop, check, report ✅!

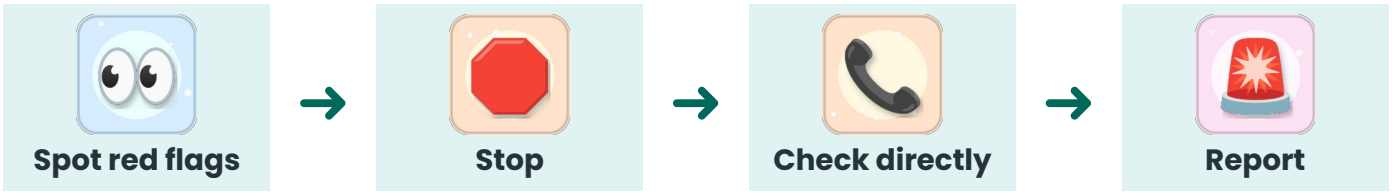
10:18

😊 Type a message...

📌 **Pinned by Bolt**

Deepfakes are AI-made fakes of real people's faces or voices. Spot red flags, stop, verify directly and report.

🔄 How it works



INDIA SPOTLIGHT

Indians can report cybercrimes, including deepfake scams, at cybercrime.gov.in or by calling the 1930 helpline.

AI CHRONICLES: The Voice That Wasn't

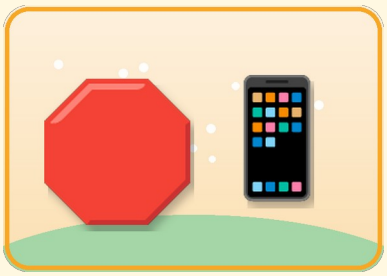
Riya | Aunt Usha | Cyber Cell

SCENE 1

Aunt Usha got a call: her son's voice crying, "Mum, I'm in trouble! Send ₹50,000 now!"

Riya noticed red flags: an unknown number, extreme urgency and odd pauses in the voice.

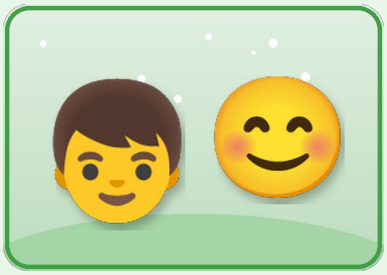
SCENE 2



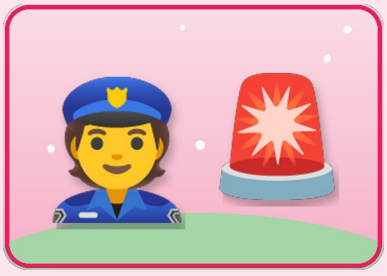
SCENE 3

"Hang up and call him back on his own number," Riya said. Aunt Usha did ☎️.

Her son answered from college, perfectly fine. The call was a voice-clone deepfake 🎭!



SCENE 4



SCENE 5

They reported it at cybercrime.gov.in and called 1930. The cyber cell thanked them.

The family made a secret code word for emergencies. "Spot, stop, check, report!"

SCENE 6

🔮 What happens next? Tick ✓ one!

☐	☐	☐
Riya teaches the code-word idea to neighbours	**Aunt Usha sends the money next time**	**The phone learns to lie**

🧐 Story check: answer the questions.

1. What red flags did Riya notice?

2. How did they verify the call?

3. What safety idea did the family adopt?

Rate this story: shade the stars.

☆ ☆ ☆ ☆ ☆

LOOK AROUND YOU: How deepfakes are misused

Fake voice asking for money

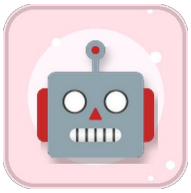
Fake news videos

Fake celebrity ads

Fake images to bully

Fake speeches

Fake investment offers



BOLT SAYS:

If a video makes you very angry or very excited, pause and check before sharing!

FUN FACT!

Scammers have used AI-cloned voices to pretend to be family members. Always call back to check!

I spy AI! Draw one AI helper you saw today.

LET'S DO IT! Deepfake Detective Training

You need:

Sample images

Checklist

Teams

Steps:

1

Teacher shows real and AI-made images.

2

Use the checklist: hands, eyes, text, lighting.

3

Vote: real or AI-made?

4

Discuss: what to do with a harmful deepfake?

THINK & TALK

Why is it harder to spot deepfakes every year?

Draw what you did!

WORKSHEET: Deepfake Alert! (Part A)

Name: _____ Date: _____

1. Circle the deepfake warning signs.

Lips don't match

Trusted source

Odd blinking

Robotic voice

Clear date & author

Strange lighting

2. If you see a harmful deepfake. Write 1, 2, 3, 4.

Report it ☐

Stop: don't share ☐

Tell a trusted adult ☐

Check the source ☐

WORKSHEET: Deepfake Alert! (Part B)

Name: _____ Date: _____

1. Red flag or OK? Tick ✓.

	Red flag	OK
"Urgent! Send money now!"		
News from a trusted source		
Lips don't match voice		
Clear date and author		
"Invest ₹100, get ₹1 lakh!"		

2. Fill in the blanks. Use the word box.

WORD BOX: deepfake ✿ report ✿ red flag

AI-made fake media is a _____.

A warning sign is a _____.

I _____ harmful deepfakes.

PUZZLE FUN: Deepfake Alert!

Name: _____ Date: _____

1. Odd one out! Circle it and justify your choice.

A				
B				
C				

2. Define it! Write each term in your own words with an example.

Deepfake

Meaning: _____

Example: _____

Red flag

Meaning: _____

Example: _____

Report

Meaning: _____

Example: _____

🧪 DEMO LAB: Deepfake red-flag cards + quiz (with teacher)

Name: _____ Date: _____

🎯 **Goal:** Practise spotting deepfakes and scam messages.

🔗 **Open:** printed cards / teacher's slides | 👩‍🏫 **With a teacher or parent**

👣 Steps

1		Teacher shows 8 images or clips: some real, some AI.
2		For each, list red flags you notice.
3		Vote: Real or AI? Then reveal answers.
4		Role-play a voice-clone scam call. Practise 'hang up and call back'.
5		Find the steps to report on cybercrime.gov.in.

🔬 Record your observations

My score out of 8:	
Best red flag:	
Reporting steps found:	

💭 **Reflect:** How can you help elders in your family stay safe from deepfake scams?

? QUIZ TIME: Deepfake Alert!

Circle the right answer!

Q1. AI-made fake media of real people is a...

a) Deepfake

b) Cake

c) Balloon

Q2. A warning sign is a...

a) Gift

b) Red flag

c) Song

Q3. A fake family voice asks for money. You...

a) Pay fast

b) Share it

c) Call back to check

Q4. Lips not matching voice is a...

a) Red flag

b) Good sign

c) Party

Q5. Harmful deepfakes should be...

a) Forwarded

b) Reported

c) Liked

MY SCORE: ____ / 5. Shade one star per correct answer.

☆ ☆ ☆ ☆ ☆

✅ TRUE OR FALSE: Deepfake Alert!

Look at the picture. Tick ✓ TRUE or FALSE.

		👍 TRUE	👎 FALSE
	Deepfakes copy real people's faces or voices.		
	Always call back to check urgent money calls.		
	Share shocking videos quickly.		
	Mismatched lips can be a red flag.		
	Deepfakes are always easy to spot.		
	Harmful deepfakes should be reported.		

HOME CONNECT

Agree on a family "safe word" to check if an urgent call is really from family.

🔢 Self-assessment: tick one.

I know it well!

I know some.

I need practice.

EXERCISE: Deepfake Alert!

Name: _____ Date: _____

A. Tick (✓) the correct answer.

1. An AI-made fake of a real person is a ____.

(a) deepfake ☐ **(b)** deep sea ☐ **(c)** headline ☐

2. Urgent requests for money are a ____.

(a) green flag ☐ **(b)** red flag ☐ **(c)** white flag ☐

3. India's cybercrime helpline number is ____.

(a) 100 ☐ **(b)** 108 ☐ **(c)** 1930 ☐

4. If a call sounds suspicious, ____ and call back.

(a) hang up ☐ **(b)** send money ☐ **(c)** share OTP ☐

B. Fill in the blanks. Use the word box.

clone **cybercrime** **report** **word**

1. AI can ____ voices from short clips.

2. Report at ____.gov.in.

3. Spot, stop, check, ____.

4. A family code ____ helps verify calls.

C. Write T for True and F for False.

1. Deepfakes can copy voices. [ ]

2. We should share deepfake videos to warn others. [ ]

3. Calling back on a known number is safe. [ ]

4. Urgency is a sign a call is genuine. [ ]

D. Match the following. Write the letter.

Column A		Column B
1. Deepfake	____	**(a)** Cybercrime helpline
2. Red flag	____	**(b)** AI fake of a person
3. 1930	____	**(c)** Family verification
4. Code word	____	**(d)** Warning sign

E. Answer in one word.

1. What is an AI fake of a person? _____

2. Which number reports cybercrime? _____

3. What is a warning sign called? _____

F. Answer these questions.

1. List four red flags of a deepfake scam.

2. What should you do if you see a deepfake video?

G. Case-based questions. Read and answer.

Aunt Usha got a call in her son's voice asking for ₹50,000 urgently. Riya noticed red flags; they called the son on his own number. He was safe. They reported it.

1. The call was a ____ deepfake.

(a) voice-clone ☐ **(b)** video ☐ **(c)** photo ☐

2. They verified by ____.

(a) sending money ☐ **(b)** calling the son's own number ☐ **(c)** posting online ☐

3. They reported it to ____.

(a) a cinema ☐ **(b)** a shop ☐ **(c)** the cyber cell ☐

H. THINK DEEPER

Should making harmful deepfakes be a crime? Discuss with one benefit and one challenge of such a law.

TOPIC 12 • MONTH 6 • RESPONSIBLE AI

Your Data, Your Right

LET'S LEARN

- Your personal data belongs to you.
- Apps must ask permission (consent) before using it.
- India's data protection law gives people rights over their data!

Your data rights

Consent

Know what's collected

Correct it

Delete it

MAGIC WORDS

Consent Your permission

Cookie Small data saved by websites

2FA Two-step login

💬 TECH TALK: Your Data, Your Right

💬 GAIO Squad Riya, Kabir, Bolt • online

👦 **Kabir**
A game app wants my contacts, location and camera 📱. Why?
10:02

🤖 **Bolt**
Good question! Your personal data belongs to you. Apps must ask consent ✋.
10:04 ✓✓

👧 **Riya**
Then we should only allow what's needed?
10:06

🤖 **Bolt**
Yes! A torch app doesn't need contacts 🔦. Deny extra permissions.
10:08 ✓✓

👦 **Kabir**
What are cookies on websites 🍪?
10:10

🤖 **Bolt**
Small files that track your browsing. You can choose 'Reject all'.
10:12 ✓✓

👧 **Riya**

How do I protect my accounts?

10:14

🤖 **Bolt**

Strong passwords plus 2FA 🔐: a second code to log in. And India's data law protects your rights ⚖️!

10:16 ✓✓

😊 Type a message...

📌 **Pinned by Bolt**

Your personal data belongs to you. Give consent carefully, allow only needed permissions, use 2FA and know your rights under India's data protection law.

🔄 How it works

Consent →

Minimum permissions →

Strong password + 2FA →

Know your rights

INDIA SPOTLIGHT

India's Digital Personal Data Protection Act, 2023 requires organisations to get consent and protect personal data.

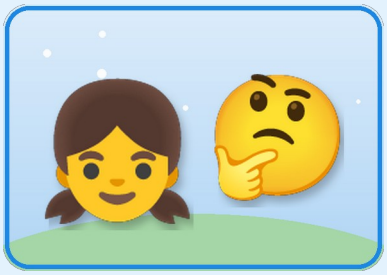
AI CHRONICLES: The Too-Curious Game

Kabir | Riya | Data Law

SCENE 1

Kabir downloaded a new puzzle game. It asked for contacts, location, microphone and camera 😳.

Riya asked, "Why does a puzzle game need your microphone?" Kabir had no answer.

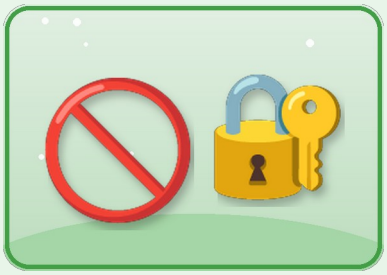


SCENE 2

SCENE 3

They read the privacy policy. The app shared data with advertisers 💰.

Kabir denied the extra permissions, deleted the game and turned on 2FA for his email.



SCENE 4

SCENE 5

In class, they learned the Data Protection Act requires apps to get clear consent ✋.

They ran a 'Permission Check Day'. 120 students cleaned up their phone settings!

SCENE 6

🔮 What happens next? Tick ✓ one!

☐	☐	☐
They make a privacy guide for parents	**Kabir allows all permissions again**	**The game apologises**

🧐 Story check: answer the questions.

1. Which permissions did the game ask for?

2. What did the privacy policy reveal?

3. What did Kabir do to stay safe?

Rate this story: shade the stars.

☆ ☆ ☆ ☆ ☆

LOOK AROUND YOU: Protect your digital rights

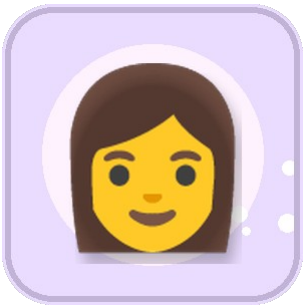
Check permissions

Read cookie pop-ups

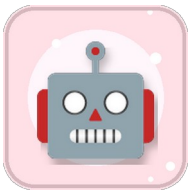
Turn off location

Read before agreeing

Use 2-step login



Ask parents first



BOLT SAYS:

"Free" apps sometimes pay their bills with your data. Read before you agree!

FUN FACT!

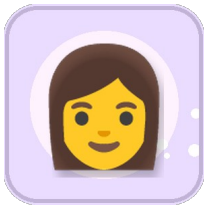
India's data protection law (2023) needs parents' consent for children's data!

👀 **I spy AI! Draw one AI helper you saw today.**

LET'S DO IT! App Permission Audit

You need:

A family phone



A parent

Audit sheet

Steps:

1		With a parent, open the settings of 3 apps.
2		List what each can access: camera, location, contacts...
3		Ask: does this app really need it?
4		Turn off what's not needed (with permission).

THINK & TALK

Which app asked for more data than it needed?

Draw what you did!

WORKSHEET: Your Data, Your Right (Part A)

Name: _____ Date: _____

1. Does the app really need it? Tick ✓.

	Needed	Not needed
Camera app → camera		
Calculator → location		
Maps → location		
Torch app → contacts		
Voice notes → microphone		

2. Fill in the blanks. Use the word box.

WORD BOX: consent ✿ 2FA ✿ cookie

Your permission is _____.

Two-step login is _____.

Small data saved by a website is a _____.

WORKSHEET: Your Data, Your Right (Part B)

Name: _____ Date: _____

1. Match the data right to its meaning.

Consent ●

● Fix wrong data

Know ●

● Give permission

Correct ●

● Remove your data

Delete ●

● See what's collected

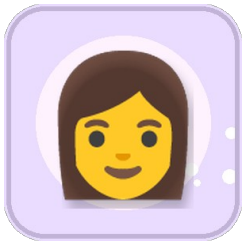
2. Circle the smart data habits.

Read before "I agree"

Location always on

Use 2FA

Give all permissions



Ask parents first

Lock your phone

PUZZLE FUN: Your Data, Your Right

Name: _____ Date: _____

1. Odd one out! Circle it and justify your choice.

A				
B				
C				

2. Define it! Write each term in your own words with an example.

Consent

Meaning: _____

Example: _____

Cookie

Meaning: _____

Example: _____

2FA

Meaning: _____

Example: _____

🧪 DEMO LAB: Your phone's privacy settings (with a parent/teacher)

Name: _____ Date: _____

🎯 **Goal:** Audit app permissions and secure an account.

🔗 **Open:** Settings → Apps → Permissions | 👩‍🏫 **With a teacher or parent**

👣 Steps

1. Open the permission list for 3 apps.
2. For each permission ask: is it really needed?
3. Turn off any permission that isn't needed.
4. Visit a website and find its cookie choices.
5. Check how to turn on 2FA for one account.

🔬 Record your observations

Permissions turned off:	
Most surprising permission:	
Account where 2FA is possible:	

💭 **Reflect:** What rights does the Digital Personal Data Protection Act give you?

? QUIZ TIME: Your Data, Your Right

Circle the right answer!

Q1. Your permission is called...

a) Consent

b) Cookie

c) Balloon

Q2. A calculator asking location is...

a) Needed

b) Suspicious

c) Fun

Q3. India's data law came in...

a) 1910

b) Never

c) 2023

Q4. Before clicking "I agree", you...

a) Read it

b) Click fast

c) Sleep

Q5. Two-step login is...

a) A game

b) 2FA

c) A cookie

MY SCORE: ____ / 5. Shade one star per correct answer.

☆ ☆ ☆ ☆ ☆

✅ TRUE OR FALSE: Your Data, Your Right

Look at the picture. Tick ✓ TRUE or FALSE.

		👍 TRUE	👎 FALSE
	Apps should ask for consent.		
	A calculator needs my location.		
	India has a data protection law from 2023.		
	You can ask for your data to be deleted.		
	Click "I agree" without reading.		
	2FA makes accounts safer.		

HOME CONNECT

Do an app-permission check on a family phone with your parents.

🔢 Self-assessment: tick one.

I know it well!

I know some.

I need practice.

EXERCISE: Your Data, Your Right

Name: _____ Date: _____

A. Tick (✓) the correct answer.

1. Permission to use your data is ____.

(a) consent ☐ **(b)** content ☐ **(c)** contest ☐

2. Small files that track browsing are ____.

(a) biscuits ☐ **(b)** cookies ☐ **(c)** crumbs ☐

3. A second login code adds ____ protection.

(a) 1FA ☐ **(b)** no ☐ **(c)** 2FA ☐

4. A torch app ____ need your contacts.

(a) does not ☐ **(b)** always ☐ **(c)** must ☐

B. Fill in the blanks. Use the word box.

belongs **needed** **Protection** **policy**

1. Your personal data ____ to you.

2. Allow only ____ permissions.

3. India's law is the Digital Personal Data ____ Act.

4. Read the privacy ____.

C. Write T for True and F for False.

1. Apps should ask for consent. [ ]

2. Give every app all permissions. [ ]

3. 2FA makes accounts safer. [ ]

4. Cookies can never be rejected. [ ]

D. Match the following. Write the letter.

Column A		Column B
1. Consent	____	**(a)** Two-step login
2. Cookie	____	**(b)** Your permission
3. 2FA	____	**(c)** How data is used
4. Privacy policy	____	**(d)** Tracking file

E. Answer in one word.

1. What is permission to use data? _____

2. What tracks browsing? _____

3. What adds a second login step? _____

F. Answer these questions.

1. How can you check and reduce app permissions?

2. What rights do people have under India's data protection law?

G. Case-based questions. Read and answer.

A puzzle game asked for contacts, location, microphone and camera, and its privacy policy said it shared data with advertisers. Kabir denied permissions and turned on 2FA.

1. The game asked for ____ permissions.

(a) unnecessary ☐ **(b)** no ☐ **(c)** only needed ☐

2. The policy showed data was ____.

(a) deleted ☐ **(b)** shared with advertisers ☐ **(c)** encrypted only ☐

3. Kabir protected his email with ____.

(a) a sticker ☐ **(b)** a shorter password ☐ **(c)** 2FA ☐

H. THINK DEEPER

Design a 'privacy label' for apps, like a food label. What five things should it show users?

UNIT REVIEW: AI CARE

Name: _____ Date: _____

Circle the right answer!

Q1. AI-made fake media is a...

a) Deepfake	b) Cake	c) Balloon

Q2. Your permission is...

a) Cookie	b) Consent	c) Balloon

Q3. Two-step login is...

a) A game	b) A cookie	c) 2FA

Q4. A warning sign is a...

a) Red flag	b) Gift	c) Song

Q5. Torch app asking contacts is...

a) Needed	b) Suspicious	c) Fun

Q6. Urgent money call from "family"? You...

a) Pay fast	b) Share it	c) Call back to check

WORD SEARCH: AI CARE

🔍 **Find these words. They go across → or down ↓.**

DEEPFAKE

RED FLAG

REPORT

CONSENT

COOKIE

2FA

R	U	T	I	W	C	O	D	L	C
E	M	O	2	F	A	I	E	P	D
D	G	K	D	B	R	U	E	G	M
F	R	G	S	U	E	S	P	C	A
L	M	H	W	O	P	K	F	O	M
A	U	D	C	S	O	S	A	O	G
G	D	W	I	K	R	G	K	K	N
R	H	E	W	G	T	S	E	I	A
G	F	A	L	T	W	W	K	E	M
N	S	N	C	O	N	S	E	N	T

I finished AI CARE! Teacher's sign: _____

MY PICTURE DICTIONARY (1/2)

2FA
Two-step login

Accessible
Usable by everyone

Cluster
A group of similar data

Confidence
How sure AI is

Confidential
Kept private

Consent
Your permission

Cookie
Small data saved by websites

Copyright
Creator's ownership

Debug
Find & fix errors

Decision tree
Yes/no choice map

Decompose
Break into parts

Deepfake
AI-made fake media

Diagnosis
Finding the illness

Error
A problem in code

Function
Reusable code block

GenAI
AI that creates

Goal
What you aim for

Iterate
Go through each item

MY PICTURE DICTIONARY (2/2)

Label
Mark AI content

List
Many items in one

ML
Machine learning

Output
The final choice

Pathway
Route to a career

Policy
Rules & guidelines

Portfolio
Collection of your work

Prompt
Instruction to AI

Red flag
Warning sign

Report
Tell the platform/adult

Sensor
Collects city data

Smart city
City using tech well

Storyboard
Plan of scenes

Stream
Group of subjects

Supervised
Learning with labels

Telemedicine
Doctor over phone/video

Test set
Data kept to check AI

? OLYMPIAD PRACTICE TEST • Part 1 of 4

Read each question. Circle the right answer. 20 questions in all!

Q1. Spam filters decide...

a) Spam or not

b) Song or not

c) Pizza or not

Q2. 90% confidence means...

a) Always right

b) Mostly right

c) Always wrong

Q3. Grouping shoppers without labels is...

a) Supervised

b) Singing

c) Unsupervised

Q4. fruits = ["a","b","c"]; fruits[2] is...

a) c

b) b

c) a

Q5. for i in range(3): runs...

a) 1 time

b) 3 times

c) 10 times

? OLYMPIAD PRACTICE TEST • Part 2 of 4

Q6. Missing ":" after if causes an...

a) Party

b) Gift

c) Error

Q7. average = total / ...

a) count

b) balloon

c) zero

Q8. AI eye screening helps detect...

a) Songs

b) Diabetic eye damage

c) Rainbows

Q9. Health AI must protect patient...

a) Balloons

b) Pizza

c) Privacy

Q10. Smart bins help keep cities...

a) Clean

b) Sleepy

c) Musical

? OLYMPIAD PRACTICE TEST • Part 3 of 4

Q11. Ramps and lifts make a city...

a) Blocked

b) Accessible

c) Costly

Q12. Biology can lead to a career in...

a) Animation

b) Driving

c) Medicine

Q13. A farmer with a crop app...

a) Uses AI

b) Builds AI

c) Writes AI law

Q14. Prompt recipe: Role, Task, Details and...

a) Cake

b) Format

c) Fun

Q15. Using a song without permission breaks...

a) Balloons

b) Cookies

c) Copyright

? OLYMPIAD PRACTICE TEST • Part 4 of 4

Q16. Odd blinking in a video may mean...

a) Deepfake

b) Real

c) Birthday

Q17. A family safe word helps stop...

a) Cooking

b) Voice-clone scams

c) Games

Q18. Children's data in India needs...

a) Nothing

b) A gift

c) Parents' consent

Q19. A flashlight app asking for contacts is...

a) Suspicious

b) Normal

c) Fun

Q20. Before "I agree", always...

a) Click fast

b) Read

c) Sleep

MY SCORE: I got ____ / 20. Colour a star for every 4 right!

CERTIFICATE

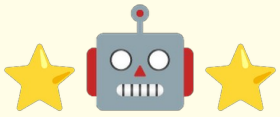
of AI Explorer

This is to say that

_____

of Class 8 finished the 6-month AI journey:

DISCOVER → CONNECT → SOLVE → RISE → CREATE → CARE



Teacher: _____ Date: _____

? ANSWER KEY (for teachers & parents)

1. How AI Makes Choices	**Quiz:** 1-a 2-b 3-c 4-a 5-b **True/False:** 1-T 2-F 3-T 4-T 5-F 6-T **Odd one out:** A: picture 4; B: picture 3; C: picture 4 **Part A 1:** Spam - Email AI; Payment - Bank AI; X-ray - Health AI; Rain - Weather AI **Part A 2:** 1-Get data, 2-Ask a question, 3-Split yes/no, 4-Reach decision **Part B 1:** a) Yes, c) Question, b) Tree **Part B 2:** decision tree, confidence, output **Exercise A:** 1-a 2-b 3-c 4-a **Exercise B:** branch, leaf, questions, 100 **Exercise C:** 1-T 2-F 3-T 4-F **Exercise D:** 1-b 2-d 3-a 4-c **Exercise E:** Decision tree; Confidence; Output **Story next:** Any sensible choice; best: The club builds a phishing-SMS checker **Exercise G (case):** 1-a 2-b 3-c **Think deeper:** Open answer; reward reasoning and examples.
2. Learning from Data	**Quiz:** 1-a 2-b 3-c 4-a 5-b **True/False:** 1-T 2-T 3-F 4-T 5-F 6-T **Odd one out:** A: picture 4; B: picture 3; C: picture 4 **Part A 1:** Photos labelled cat/dog: Supervised; Grouping shoppers by habits: Unsupervised; Emails labelled spam: Supervised; Grouping songs by mood: Unsupervised **Part A 2:** labels, cluster, test set **Part B 1:** X-rays - Supervised; Shoppers - Unsupervised; Emails - Supervised; News - Unsupervised **Part B 2:** 1-Collect data, 2-Train, 3-Test on new data, 4-Use it **Exercise A:** 1-a 2-b 3-c 4-a **Exercise B:** labelled, misleading, customers, data **Exercise C:** 1-T 2-F 3-T 4-T **Exercise D:** 1-b 2-d 3-a 4-c **Exercise E:** Supervised; Cluster; Test set **Story next:** Any sensible choice; best: Kabir builds an app for the whole village **Exercise G (case):** 1-a 2-b 3-c **Think deeper:** Open answer; reward reasoning and examples.
3. Python Playground	**Quiz:** 1-a 2-b 3-c 4-a 5-b **True/False:** 1-T 2-T 3-F 4-T 5-T 6-F **Odd one out:** A: picture 4; B: picture 3; C: picture 4 **Part A 1:** apple, 3, mango **Part A 2:** [ ] - A list; for - Loops; def - Makes a function; len() - Counts items **Part B 1:** Circle: nums = [1, 2, 3], for n in nums:, def hello(): **Part B 2:** for, def, len **Exercise A:** 1-a 2-b 3-c 4-a **Exercise B:** def, items, iterate, brackets **Exercise C:** 1-T 2-F 3-T 4-F **Exercise D:** 1-b 2-d 3-a 4-c **Exercise E:** def; len(); append() **Story next:** Any sensible choice; best: Riya adds a book-recommendation feature **Exercise G (case):** 1-a 2-b 3-c **Think deeper:** Open answer; reward reasoning and examples.

4. Code a Solution	**Quiz:** 1-a 2-b 3-c 4-a 5-b **True/False:** 1-T 2-T 3-F 4-T 5-T 6-F **Odd one out:** A: picture 4; B: picture 3; C: picture 4 **Part A 1:** 1-Understand the problem, 2-Break it down, 3-Write code, 4-Debug & improve **Part A 2:** print("Hello"): Correct; print("Hello": Has a bug; nums = [1, 2, 3]: Correct; if x > 5 print(x): Has a bug; for n in nums:: Correct **Part B 1:** Average - sum / len; Quiz - input + if; Fever - if temp > 100; Tables - for loop 1 to 10 **Part B 2:**), :, " **Exercise A:** 1-a 2-b 3-c 4-a **Exercise B:** len, if, clues, improve **Exercise C:** 1-T 2-F 3-T 4-F **Exercise D:** 1-b 2-d 3-a 4-c **Exercise E:** Debugging; Decomposition; elif **Story next:** Any sensible choice; best: The school uses their slot system daily **Exercise G (case):** 1-a 2-b 3-c **Think deeper:** Open answer; reward reasoning and examples.
5. AI for Better Health	**Quiz:** 1-a 2-b 3-c 4-a 5-b **True/False:** 1-T 2-F 3-T 4-T 5-F 6-T **Odd one out:** A: picture 4; B: picture 3; C: picture 4 **Part A 1:** Eye scanner - Checks diabetic eyes; Telemedicine - Helps villages; Smart watch - Tracks heart rate; X-ray AI - Spots TB signs **Part A 2:** diagnosis, telemedicine, confidential **Part B 1:** Scan 1000 X-rays fast: AI helps; Choose final treatment: Doctor decides; Medicine reminders: AI helps; Comfort a patient: Doctor decides **Part B 2:** Circle: Medical reports, Medicines taken, Heart rate, Blood group **Exercise A:** 1-a 2-b 3-c 4-a **Exercise B:** scans, decide, eSanjeevani, diagnosis **Exercise C:** 1-T 2-F 3-T 4-F **Exercise D:** 1-b 2-d 3-a 4-c **Exercise E:** Doctor; Telemedicine; Yes **Story next:** Any sensible choice; best: Riya volunteers at the next health camp **Exercise G (case):** 1-a 2-b 3-c **Think deeper:** Open answer; reward reasoning and examples.
6. AI for Better Cities	**Quiz:** 1-a 2-b 3-c 4-a 5-b **True/False:** 1-T 2-T 3-F 4-T 5-T 6-F **Odd one out:** A: picture 4; B: picture 3; C: picture 4 **Part A 1:** Traffic - Smart signals; Bins - Bins that alert; Leaks - Leak sensors; Street light - Lights dim when empty **Part A 2:** Child's own work **Part B 1:** Circle: Smart signals, Bins that alert, Auto street lights, Air quality boards **Part B 2:** sensor, accessible, smart city **Exercise A:** 1-a 2-b 3-c 4-a **Exercise B:** AI, wheelchair, see, Smart **Exercise C:** 1-T 2-F 3-T 4-F **Exercise D:** 1-b 2-d 3-a 4-c **Exercise E:** Sensor; Accessible; Smart Cities Mission **Story next:** Any sensible choice; best: Kabir's team designs smart bus stops **Exercise G (case):** 1-a 2-b 3-c **Think deeper:** Open answer; reward reasoning and examples.

7. Find Your Future Path	**Quiz:** 1-a 2-b 3-c 4-a 5-b **True/False:** 1-T 2-T 3-F 4-T 5-T 6-F **Odd one out:** A: picture 4; B: picture 3; C: picture 4 **Part A 1:** Maths - Data scientist; Biology - Doctor; Art - Animator; Computer Science - AI engineer **Part A 2:** pathway, stream, goal **Part B 1:** Child's own answer **Part B 2:** Child's own work **Exercise A:** 1-a 2-b 3-c 4-a **Exercise B:** strengths, goal, science, adjust **Exercise C:** 1-T 2-F 3-T 4-F **Exercise D:** 1-b 2-d 3-a 4-c **Exercise E:** Stream; Pathway; Bioinformatics **Story next:** Any sensible choice; best: Riya builds an AI health-app prototype **Exercise G (case):** 1-a 2-b 3-c **Think deeper:** Open answer; reward reasoning and examples.
8. AI Career Discovery	**Quiz:** 1-a 2-b 3-c 4-a 5-b **True/False:** 1-T 2-T 3-F 4-T 5-T 6-F **Odd one out:** A: picture 4; B: picture 3; C: picture 4 **Part A 1:** ML engineer: Builds AI; Doctor with AI scanner: Uses AI; Robotics engineer: Builds AI; Farmer with crop app: Uses AI; AI data engineer: Builds AI **Part A 2:** Child's own work **Part B 1:** ML engineer - Builds AI; Farmer - Uses AI; Policy expert - Guides AI; AI teacher - Teaches AI **Part B 2:** ML, portfolio, policy **Exercise A:** 1-a 2-b 3-c 4-a **Exercise B:** learning, insights, fair, IndiaAI **Exercise C:** 1-T 2-F 3-T 4-F **Exercise D:** 1-b 2-d 3-a 4-c **Exercise E:** Machine learning; Portfolio; Policy expert **Story next:** Any sensible choice; best: Kabir mentors juniors on portfolios **Exercise G (case):** 1-a 2-b 3-c **Think deeper:** Open answer; reward reasoning and examples.
9. Build with GenAI	**Quiz:** 1-a 2-b 3-c 4-a 5-b **True/False:** 1-T 2-T 3-F 4-T 5-T 6-F **Odd one out:** A: picture 4; B: picture 3; C: picture 4 **Part A 1:** Role - You are a coach; Task - Make a plan; Details - for Class 8, 2 weeks; Format - as a table **Part A 2:** As a teacher, make 5 MCQs on plants: Strong; Do something: Weak; Explain gravity in 3 bullet points: Strong; Write stuff: Weak **Part B 1:** 1-Plan your idea, 2-Write the prompt, 3-Check the output, 4-Improve & iterate **Part B 2:** Child's own work **Exercise A:** 1-a 2-b 3-c 4-a **Exercise B:** prompt, textbook, bugs, better **Exercise C:** 1-T 2-F 3-T 4-F **Exercise D:** 1-b 2-d 3-a 4-c **Exercise E:** GenAI; Prompt; Iterating **Story next:** Any sensible choice; best: They add a leaderboard feature **Exercise G (case):** 1-a 2-b 3-c **Think deeper:** Open answer; reward reasoning and examples.

10. AI Media Studio	**Quiz:** 1-a 2-b 3-c 4-a 5-b **True/False:** 1-T 2-T 3-F 4-T 5-F 6-T **Odd one out:** A: picture 4; B: picture 3; C: picture 4 **Part A 1:** 1-Storyboard, 2-Shoot, 3-Edit, 4-Label & share **Part A 2:** storyboard, copyright, label **Part B 1:** Video editor - Cuts clips; Audio cleaner - Removes noise; Auto subtitles - Adds text; Image maker - Creates pictures **Part B 2:** Labelling AI images: Ethical; Using songs without permission: Not ethical; Adding subtitles for all: Ethical; Faking a friend's voice: Not ethical **Exercise A:** 1-a 2-b 3-c 4-a **Exercise B:** royalty, subtitles, real, creators **Exercise C:** 1-T 2-F 3-T 4-F **Exercise D:** 1-b 2-d 3-a 4-c **Exercise E:** Storyboard; Copyright; Label **Story next:** Any sensible choice; best: The team makes a film on plastic waste **Exercise G (case):** 1-a 2-b 3-c **Think deeper:** Open answer; reward reasoning and examples.
11. Deepfake Alert!	**Quiz:** 1-a 2-b 3-c 4-a 5-b **True/False:** 1-T 2-T 3-F 4-T 5-F 6-T **Odd one out:** A: picture 4; B: picture 3; C: picture 4 **Part A 1:** Circle: Lips don't match, Odd blinking, Robotic voice, Strange lighting **Part A 2:** 1-Stop: don't share, 2-Check the source, 3-Report it, 4-Tell a trusted adult **Part B 1:** "Urgent! Send money now!": Red flag; News from a trusted source: OK; Lips don't match voice: Red flag; Clear date and author: OK; "Invest ₹100, get ₹1 lakh!": Red flag **Part B 2:** deepfake, red flag, report **Exercise A:** 1-a 2-b 3-c 4-a **Exercise B:** clone, cybercrime, report, word **Exercise C:** 1-T 2-F 3-T 4-F **Exercise D:** 1-b 2-d 3-a 4-c **Exercise E:** Deepfake; 1930; Red flag **Story next:** Any sensible choice; best: Riya teaches the code-word idea to neighbours **Exercise G (case):** 1-a 2-b 3-c **Think deeper:** Open answer; reward reasoning and examples.
12. Your Data, Your Right	**Quiz:** 1-a 2-b 3-c 4-a 5-b **True/False:** 1-T 2-F 3-T 4-T 5-F 6-T **Odd one out:** A: picture 4; B: picture 3; C: picture 4 **Part A 1:** Camera app → camera: Needed; Calculator → location: Not needed; Maps → location: Needed; Torch app → contacts: Not needed; Voice notes → microphone: Needed **Part A 2:** consent, 2FA, cookie **Part B 1:** Consent - Give permission; Know - See what's collected; Correct - Fix wrong data; Delete - Remove your data **Part B 2:** Circle: Read before "I agree", Use 2FA, Ask parents first, Lock your phone **Exercise A:** 1-a 2-b 3-c 4-a **Exercise B:** belongs, needed, Protection, policy **Exercise C:** 1-T 2-F 3-T 4-F **Exercise D:** 1-b 2-d 3-a 4-c **Exercise E:** Consent; Cookie; 2FA **Story next:** Any sensible choice; best: They make a privacy guide for parents **Exercise G (case):** 1-a 2-b 3-c **Think deeper:** Open answer; reward reasoning and examples.

Unit reviews & practice test

AI DISCOVER: 1-a 2-b 3-c 4-a 5-b 6-c • AI CONNECT: 1-a 2-b 3-c 4-a 5-b 6-c • AI SOLVE: 1-a 2-b 3-c 4-a 5-b 6-c • AI RISE: 1-a 2-b 3-c 4-a 5-b 6-c • AI CREATE: 1-a 2-b 3-c 4-a 5-b 6-c • AI CARE: 1-a 2-b 3-c 4-a 5-b 6-c

Olympiad Practice Test: 1-a 2-b 3-c 4-a 5-b 6-c 7-a 8-b 9-c 10-a 11-b 12-c 13-a 14-b 15-c 16-a 17-b 18-c 19-a 20-b

WORD SEARCH ANSWERS

H N O L A B E L S F
C O N F I D E N C E
M T M C P S D F S N
T E S T S E T M R A
R B K W U U N F F S
H A G T T H N S M U
M P I C L U S T E R
S U P E R V I S E D
O U T P U T T W A N
S E S T G O B R M U

AI DISCOVER

M D P F N P S H R I
R E S S M P P M U T
P C F U N C T I O N
R O H L F W I I R K
K M S T S D T S W U
O P K G R E E S E M
W O C L A B R G R D
B S U B I U A U R H
D E S E I G T H O G
B O L I S T E B R B

AI CONNECT

N U S E N S O R P E
M D B E R G I O K O
S P R I V A C Y N U
M T U O U H L A I W
F L T U U D G U I K
D C R R C M C O E A
K O O D O C T O R D
B S M A R T C I T Y
A C C E S S I B L E
W D I A G N O S I S

AI SOLVE

G O P O L I C Y O W
K P O R T F O L I O
O G O A L P F H K I
B C B P I S T R L E
G C O G P I F M M O
U L S T R E A M L T
G L D B H I U W H D
L F K P A B M C K L
A L P A T H W A Y K
L E O W C K W G P K

AI RISE

F C P E I C L K S W
U O R A W L C K T M
K P O R L F R R O F
B Y M G E N A I R I
A R P M N A T O Y M
N I T L A B E L B U
A G P B F W G D O H
P H M S M S I P A D
U T M K B O C G R L
S I T E R A T E D W

AI CREATE

R U T I W C O D L C
E M O 2 F A I E P D
D G K D B R U E G M
F R G S U E S P C A
L M H W O P K F O M
A U D C S O S A O G
G D W I K R G K K N
R H E W G T S E I A
G F A L T W W K E M
N S N C O N S E N T

AI CARE

MY NOTES

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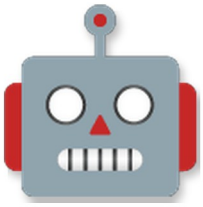

About this book

A hands-on AI skills book for senior learners of Class 8, following the 6-month Global AI Olympiad journey.

- 💬 Tech Talk: concepts as a group chat
- 🧪 Demo Lab: hands-on real AI tools
- 📚 AI Chronicles: 12 illustrated stories
- 📝 Standard, case-based & HOTS questions
- 🏆 Olympiad practice test & certificate

AI Basics **Communicate with AI** **AI Applications**

AI & Career **AI Tools** **Responsible AI**

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